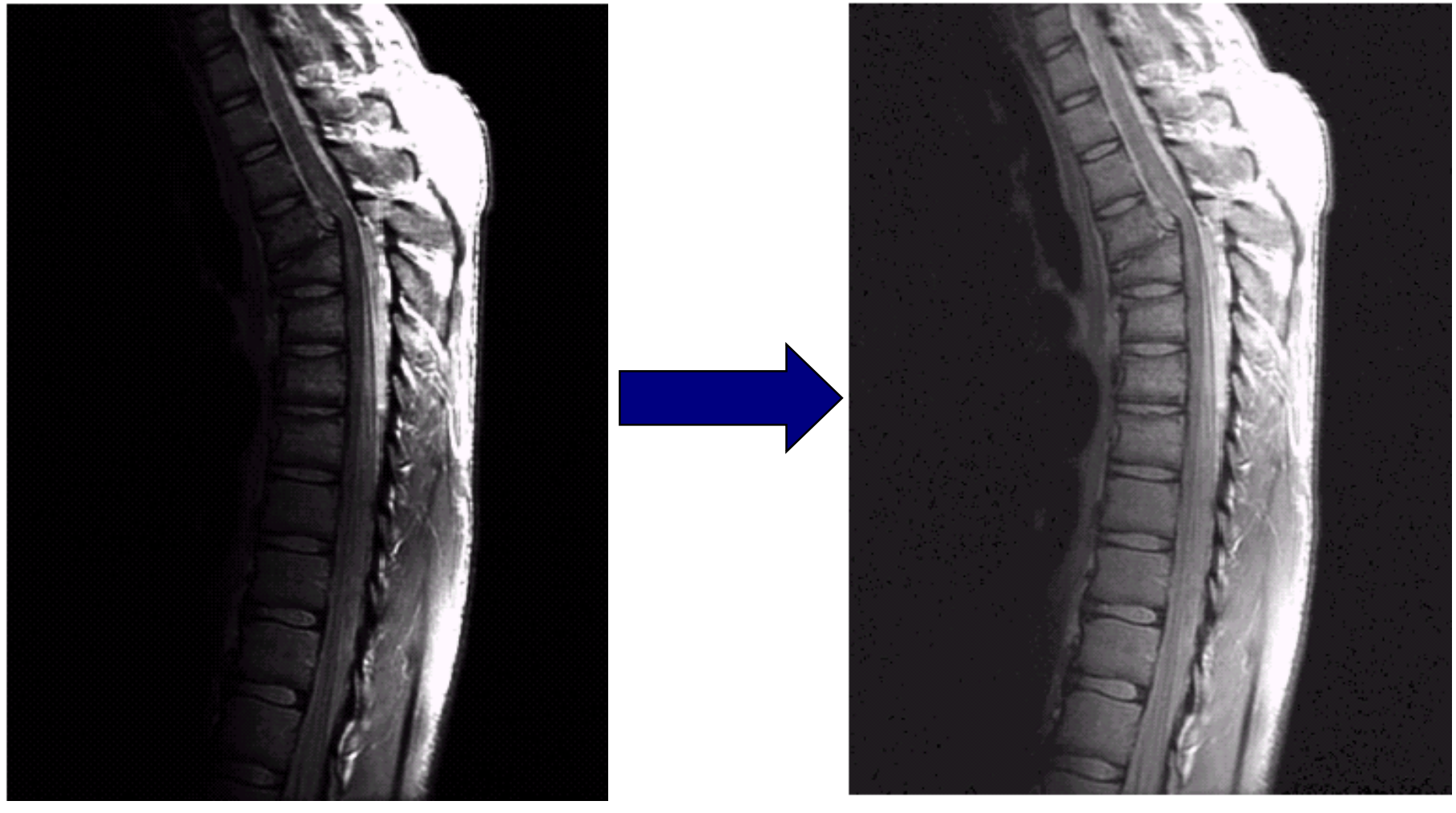
Image Enhancement and filter in Spatial Domain

UNIT - 2

Image Enhancement

- Image enhancement refers to the sharpening or improving the quality of an image by enhancement of pixel of point in a spatial domain or frequency domain.
- It the done to make image more useful.
- The reasons for doing this include:
  - Highlighting interesting detail in images
  - Removing noise from images
  - Making images more visually appealing



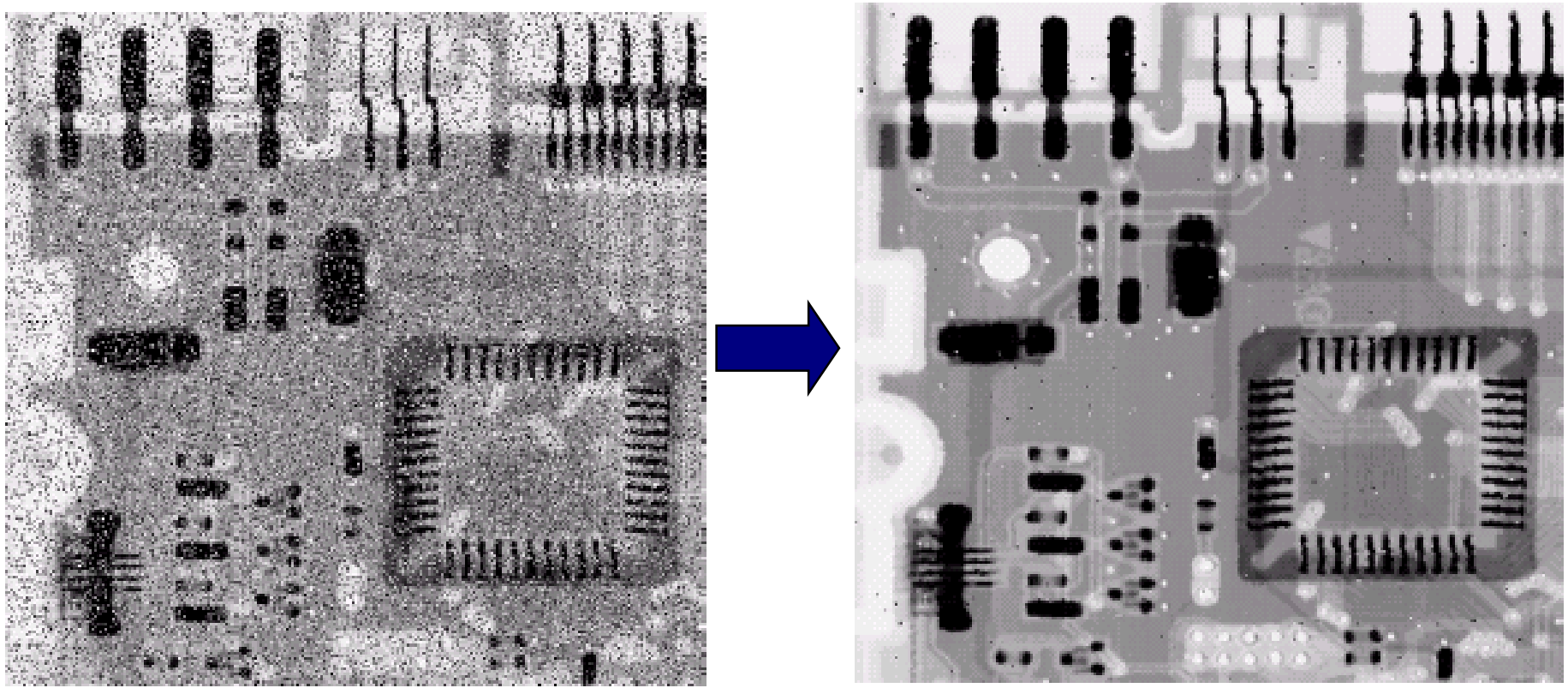


Image Enhancement Techniques

- There are two broad categories of image enhancement techniques

1. Spatial domain technique
   - Direct manipulation of image pixels
2. Frequency domain techniques
   - Manipulation of Fourier transform or wavelet transform of image.

Basic Spatial Domain Image Enhancement

- Most spatial domain enhancement operations can be denoted by the expression

$$g(x, y) = T(f(x, y))$$

where,

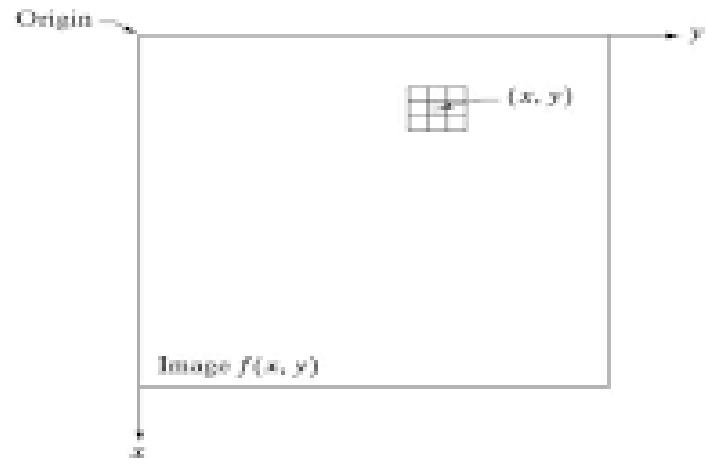
$f(x, y)$ is the input image,

$g(x, y)$ is the output image, and

T is an operator on f defined over some neighborhood of (x, y) .

- The operator can apply to a single image or to a set of images.

A 3 x 3 neighborhood about a point (x, y) in an image



Gray Level Transformation

- Some basic gray level transformation in needed to enhance the image in a spatial domain.
- Some basic gray level transformation is listed as below:
  1. Point operation
  2. Contrast stretching
  3. Thresholding
  4. Digital negative
  5. Intensity level slicing (Gray Level Slicing)
  6. Bit plane Slicing

Point Operation/Processing

- Point operation is *defined as a function that are performed on each pixel of an image*. i.e. *g depends only on the value of f at a single point (x ,y)* and T becomes **intensity (also called gray-level or mapping) transformation function** of the form
- s = T (r), where, s and r denotes the intensity of g and f at any point (x, y).
- Point operation perform a modification of the pixel without changing the size, geometry, or local structure of image.
- This operation can be used in many situation and for different purpose such as
  - modifying image brightness or contrast (contrast stretching),
  - applying arbitrary intensity transformation (“curves”)
  - quantizing (or “posterizing”) images,
  - global thresholding
  - gamma transformations

Some Basic Intensity (Gray Level) Transformation Function

- Intensity transformations are among the simplest of all image processing techniques.
- The pixel values, before and after processing, will be denoted by *r* and *s*, respectively. The values are related by an expression of the form $\boldsymbol{s = T(r)}$, where *T* is a transformation that maps a pixel value *r* into a pixel value *s*.
- Before we are dealing with digital quantities, values of a transformation function typically are stored in a one-dimensional (1-D) array and the mapping from r to s are implemented via **table lookups**. For an 8-bit environment, a lookup table containing the values of T will have 256 entries.
- Some basic intensity transformations are:
  1. **Linear** : negative and identity transformations
  2. **Logarithmic** : log and inverse-log transformations
  3. **Power-law** : nth power and nth root transformations

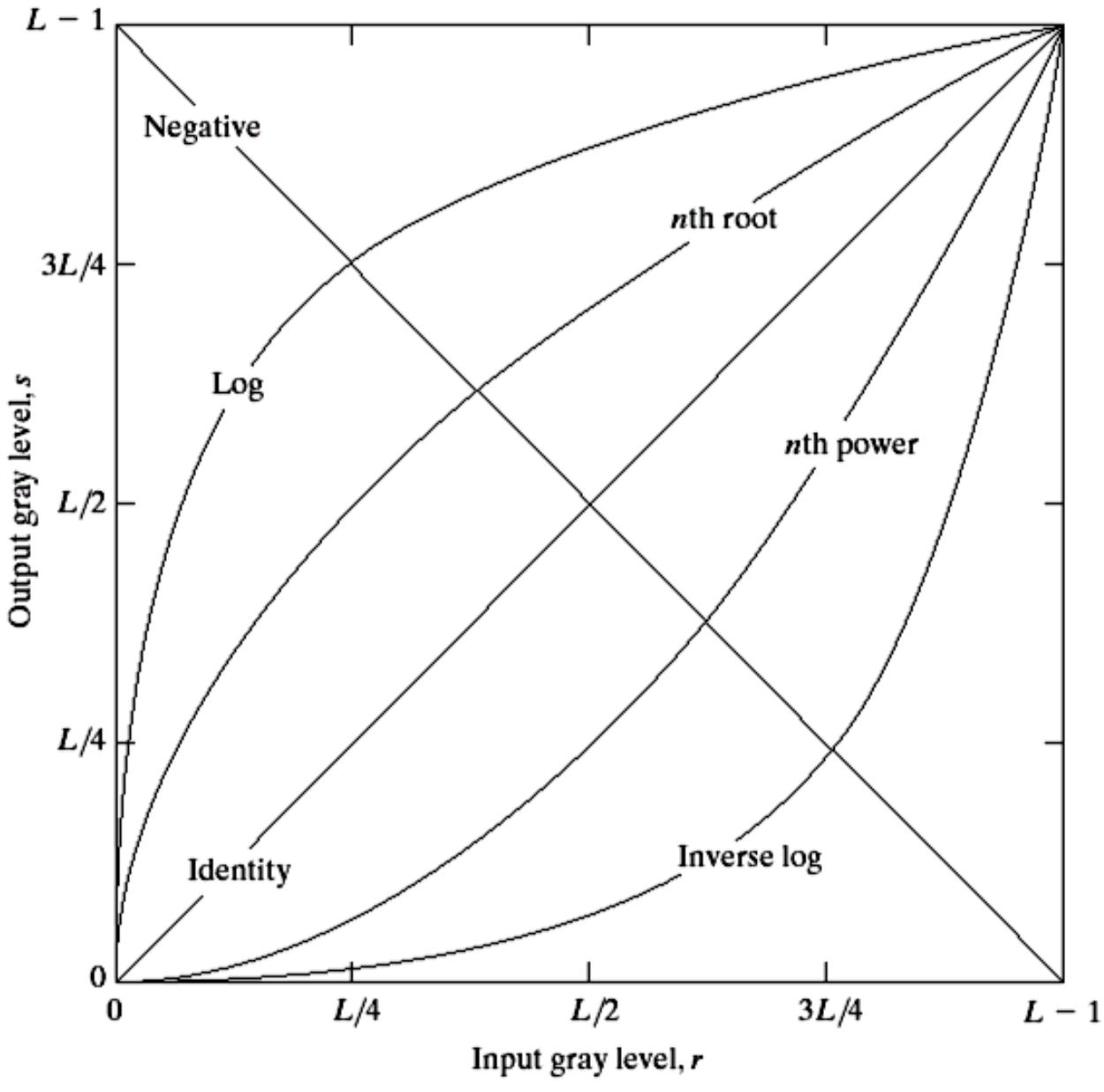


Figure: Some basic intensity transformation functions.

- The identity function is the trivial case in which output intensities are identical to input intensities.
- It is included in the graph only for completeness.

Image Negative

- As the name suggest, negative means inverting the gray level i.e. black into white and vice-versa.
- Therefore, the negative of an image with the intensity level in the range [0, L – 1] is obtained by using the negative transformation.
- It is given by $s = T(r) = L - 1 - r$
- It produce a photographic negative. Common example is x-ray image.
- This type of processing is particularly useful for enhancing white or gray detail embedded in dark regions of an image, especially when the black areas are dominant in size.
- For example:

  if r = 0, s = 256 – 1 – 0 = 255 (White)

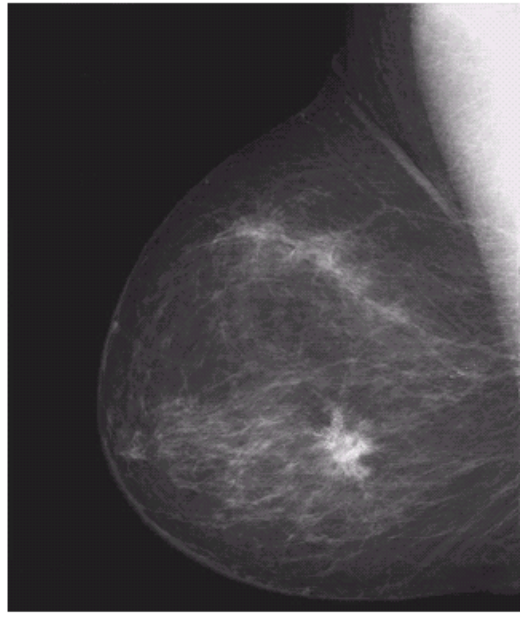
  if r = 255, s = 256 – 1 – 255 = 0 (Black)

  If r = 200 , s = 256 – 1 – 200 = 55

Example: Negative Images

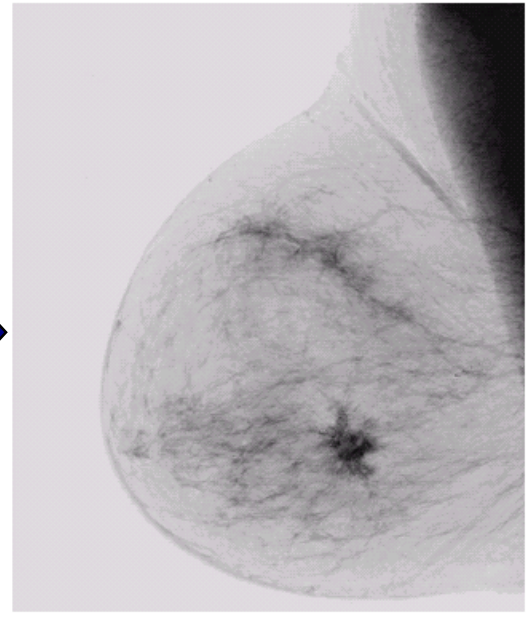
Note how much clearer the tissue is in the negative image of the mammogram ?

Original Image



(a)

$s = 1.0 - r$



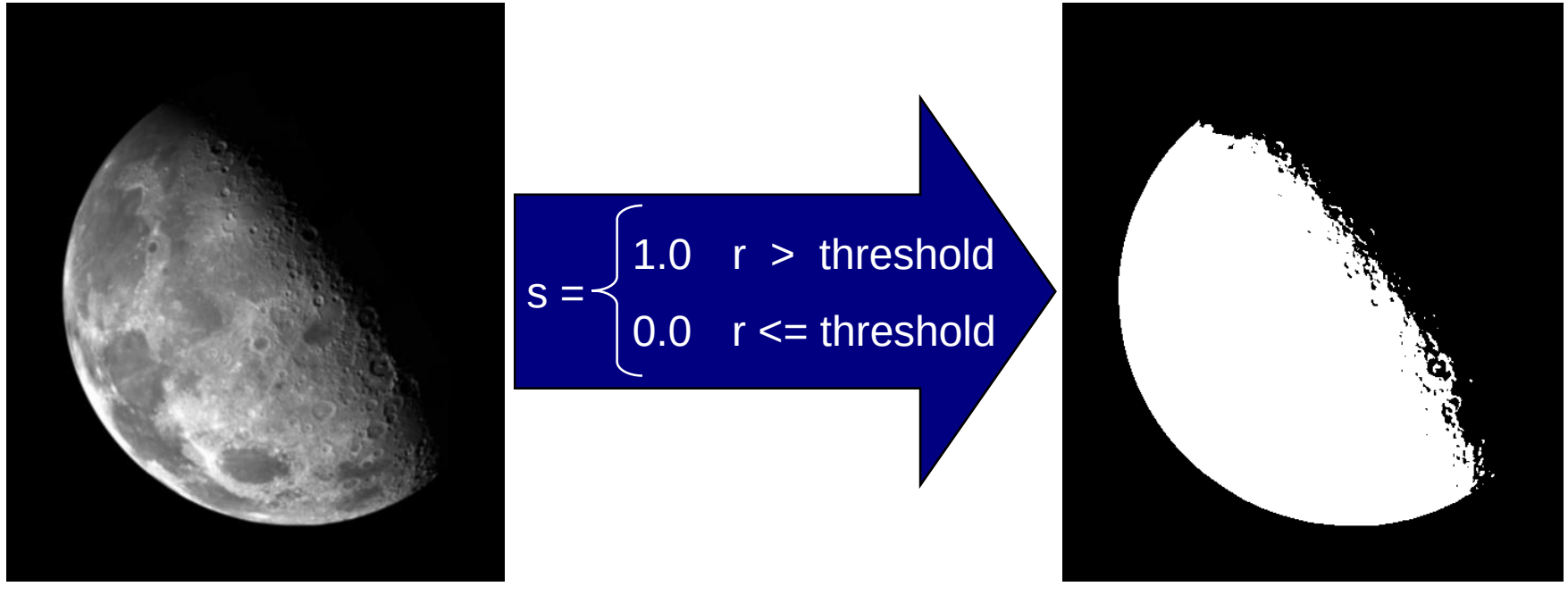
(b)

Negative Image

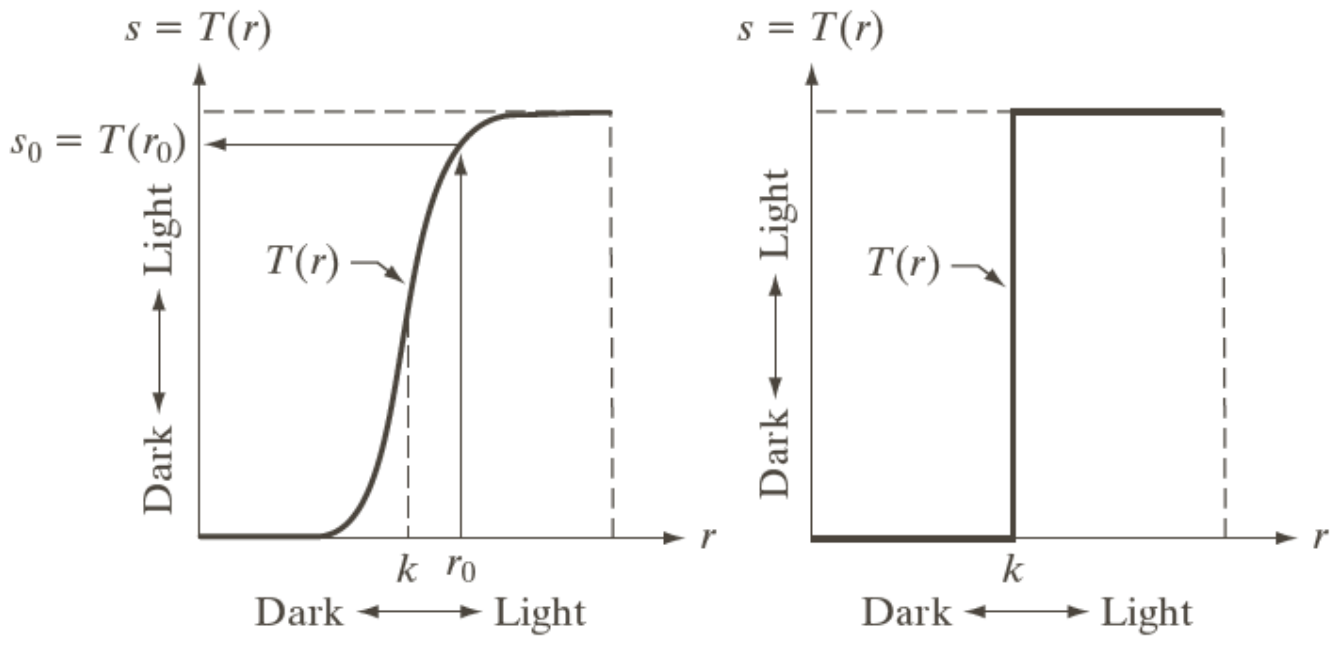
Figure: (a) Original Image mammogram, (b) Negative image obtained using the negative transformation

Thresholding

- Thresholding transformations are particularly useful for segmentation in which we want to isolate an object of interest from a background.



Intensity Transformation



Log Transformations

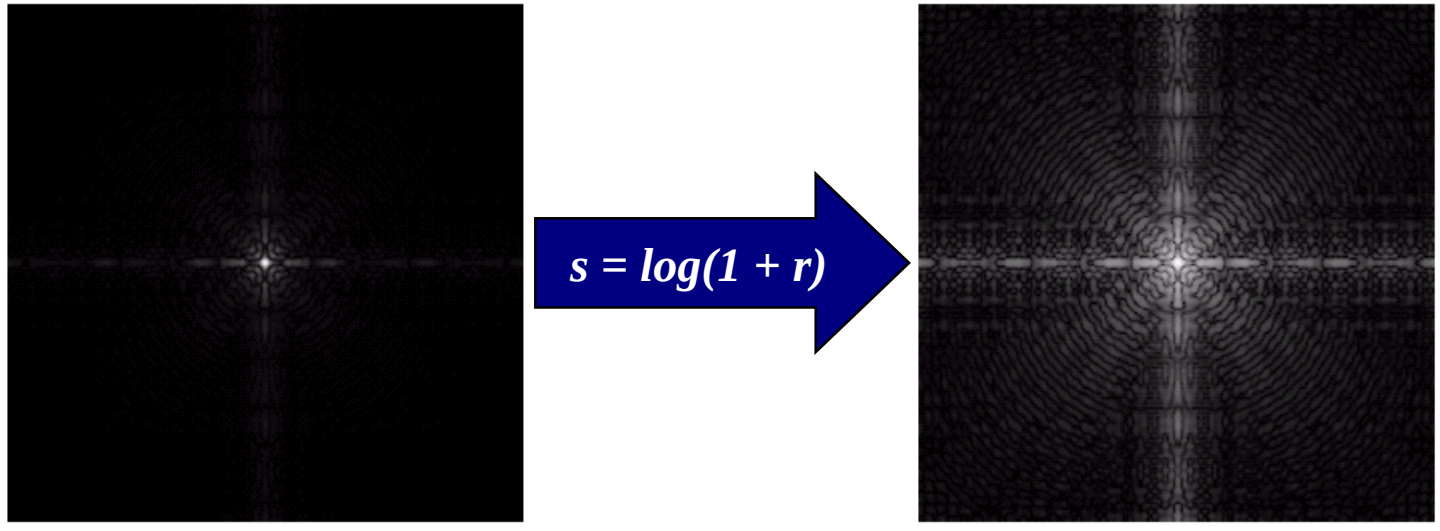
- The general form of the log transformation is:

$$s = c * \log(1 + r)$$

- Where c is a constant, and it is assumed that $r \geq 0$.
- The log transformation maps a narrow range of low input grey level values into a wider range of output values. The inverse log transformation performs the opposite transformation.
- The log function compresses the dynamic range of images with large variations in pixel values.

Log Transformation

- Log functions are particularly useful when the input grey level values may have an extremely large range of values.
- In the following example the Fourier transform of an image is put through a log transform to reveal more detail



(a) (b)

Figure: (a) Fourier spectrum. (b) Result of applying the log transformation in above equation with $c = 1$.

We usually set c to 1

Grey levels must be in the range [0.0, 1.0]

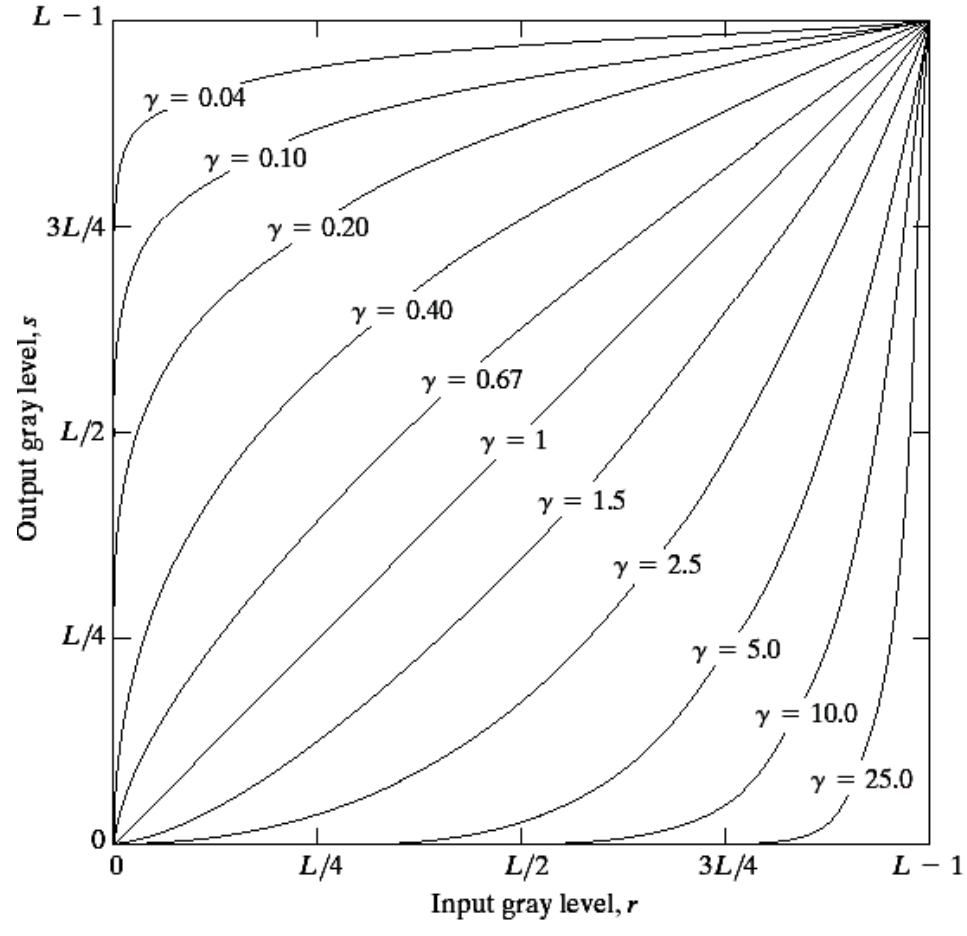
Power Law (gamma) Transformations

- Power law transformations have the basic form

$$s = c * r^{\gamma}$$

where c and γ are positive constants.

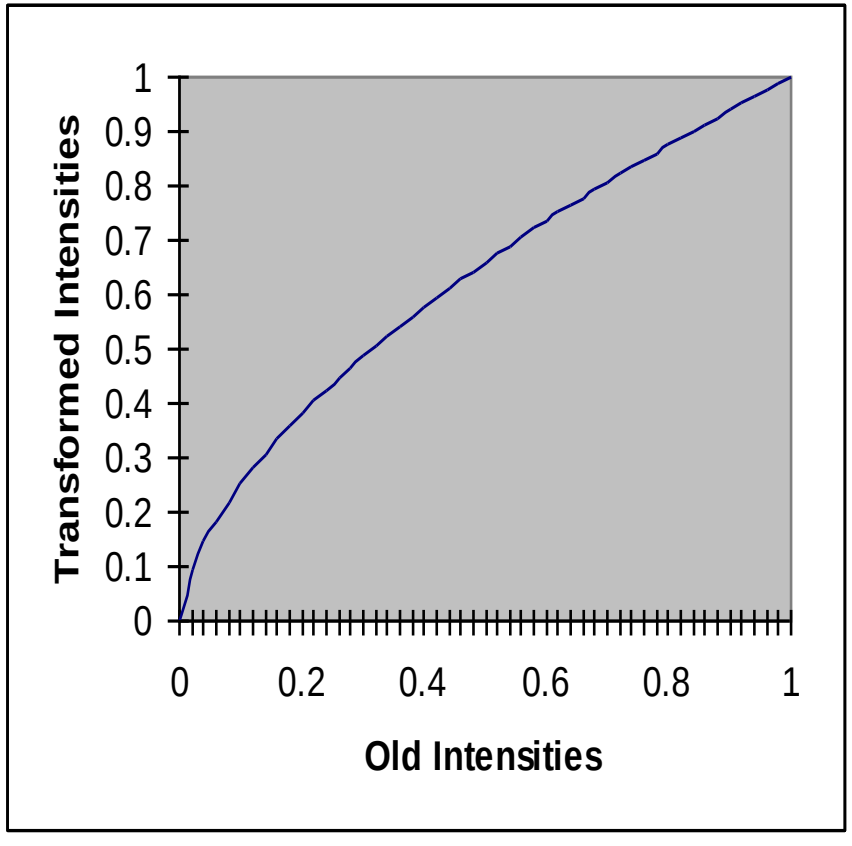
- As in case of log transformations, power law curves with fractional values of γ map a narrow range of dark input values into a wider range of output values, and vice-versa.
- Unlike log transformations, we have a family of possible transformation curves obtained by simply by varying γ.



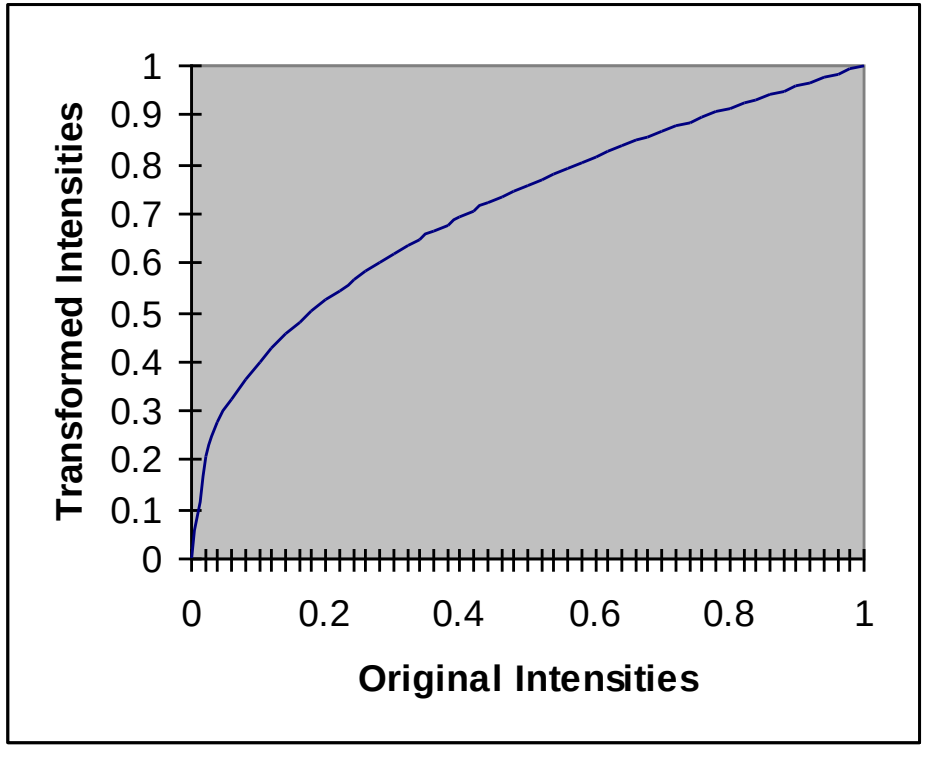
We usually set *c* to 1

Figure: Magnetic resonance image (MRI) of a fractured human spine

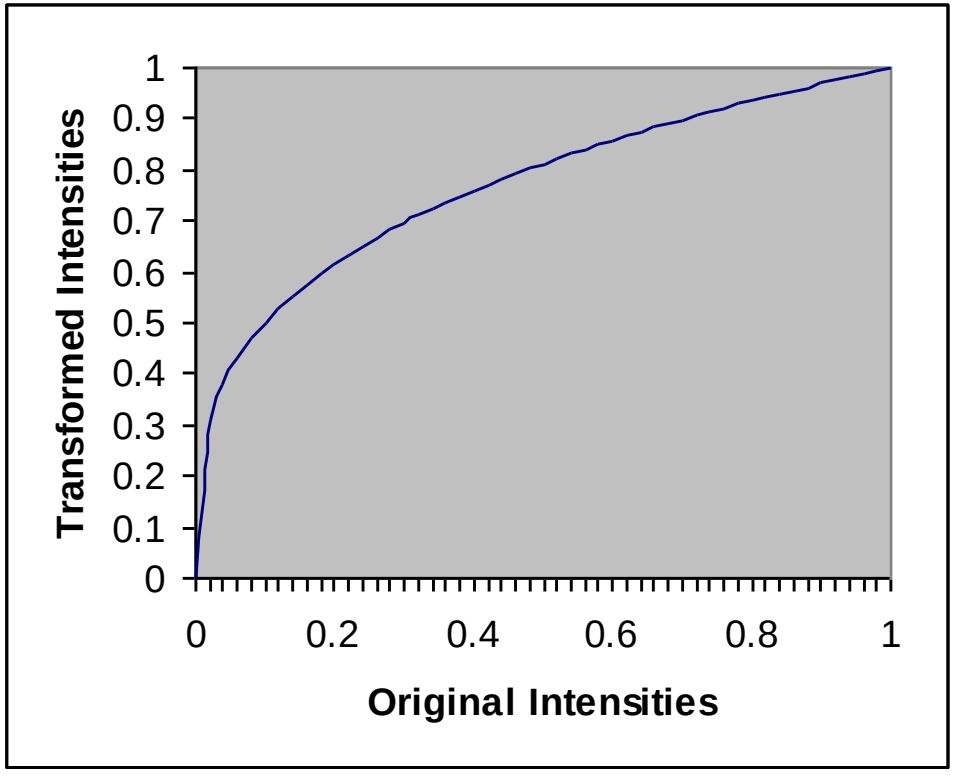
$\gamma = 0.6$


Transformed Intensities
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.2
0.4
0.6
0.8
1
Old Intensities

$\gamma = 0.4$


Transformed Intensities
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.2
0.4
0.6
0.8
1
Original Intensities

$\gamma = 0.3$


Transformed Intensities
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.2
0.4
0.6
0.8
1
Original Intensities

- The images to the right show a magnetic resonance image (MRI) of a fractured human spine.
- Different curves highlight different details.

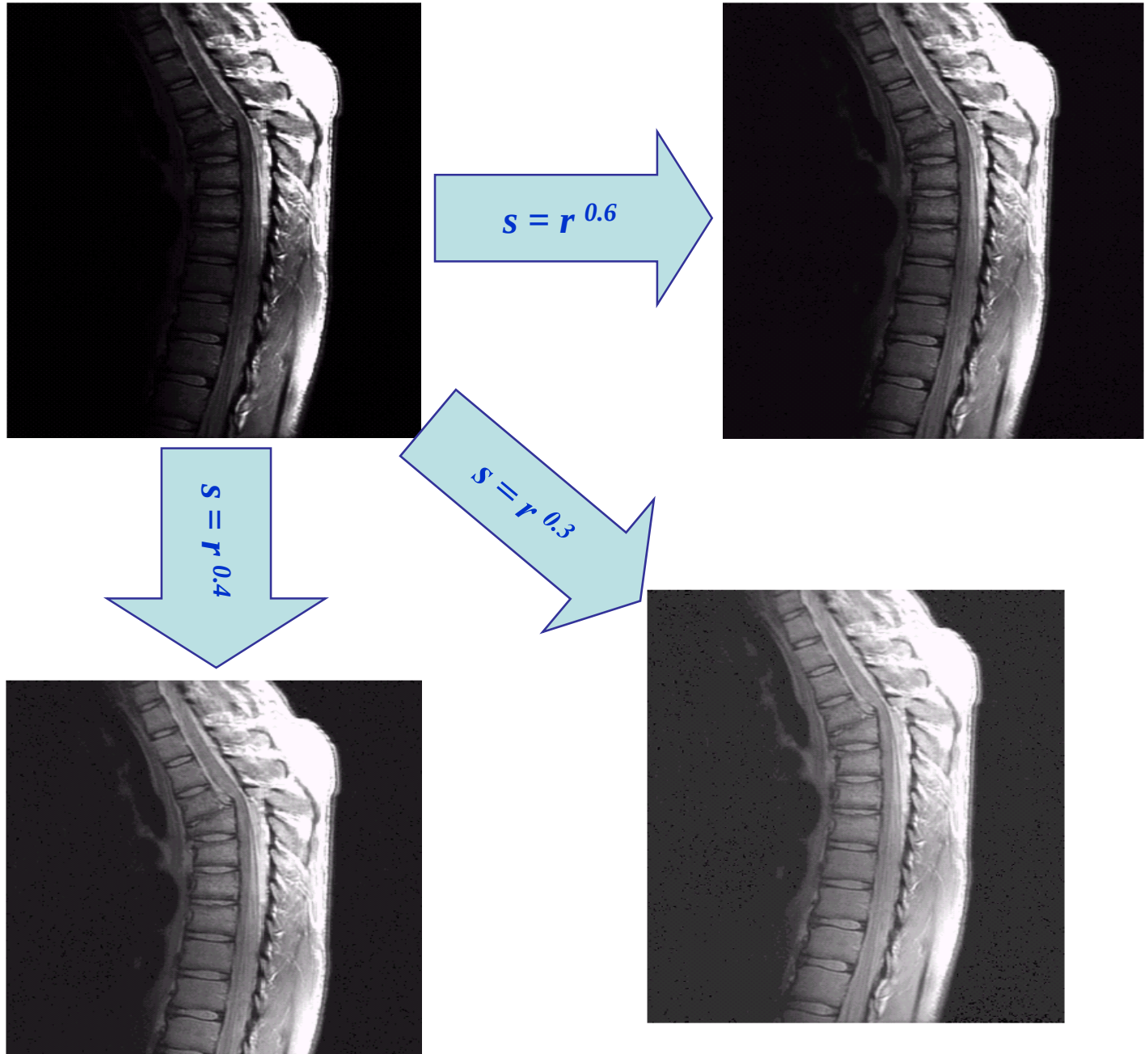
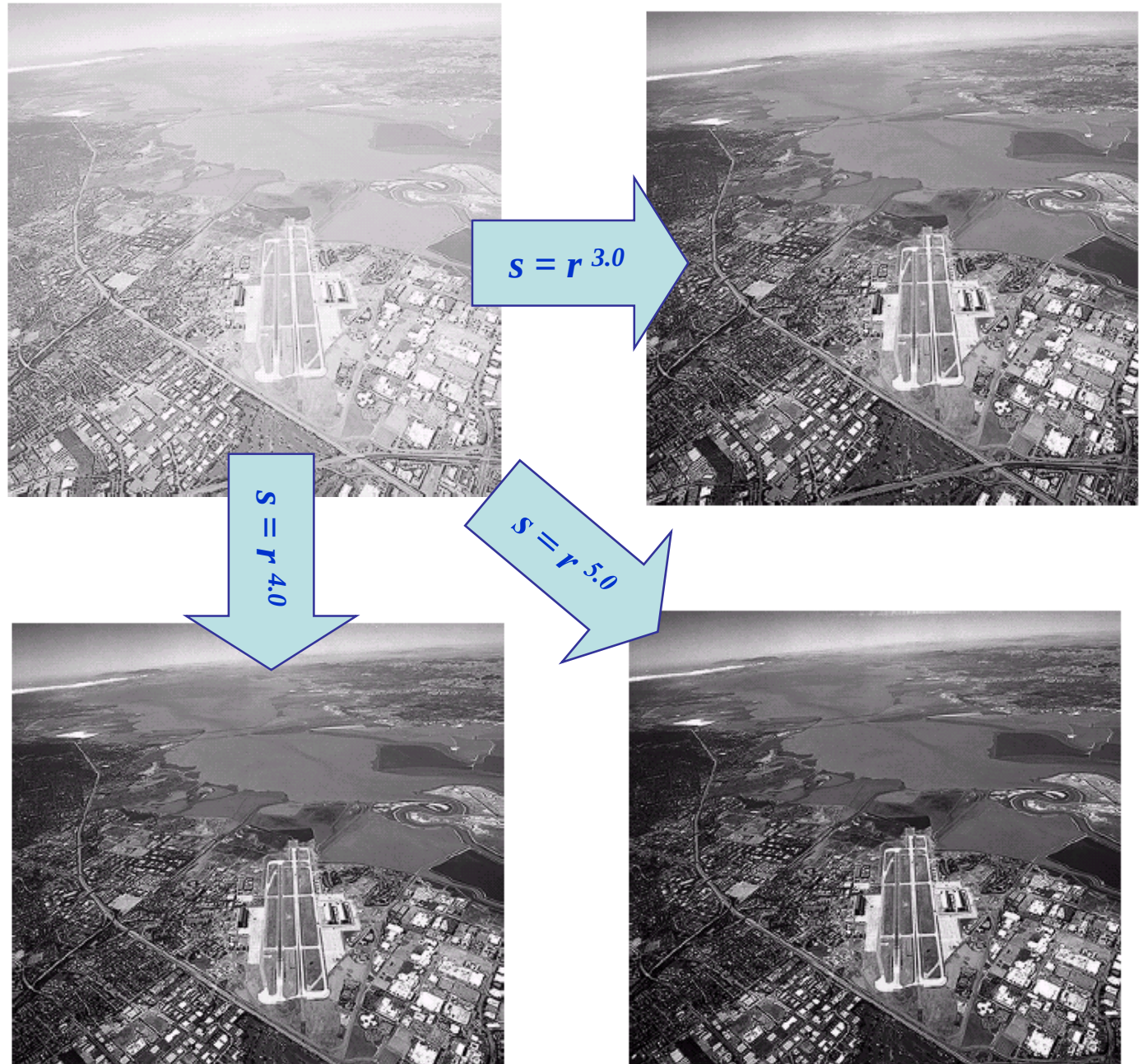


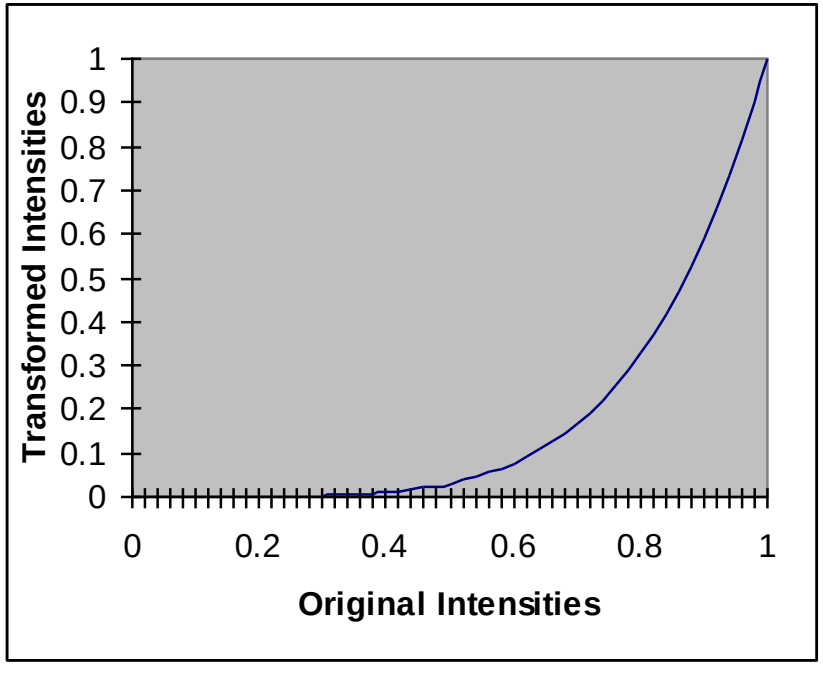
Figure: An aerial photo of a runway is shown.

- This time power law are used to darken the image.
- Different curves highlight different detail.



Power Law Example

$\gamma = 5.0$

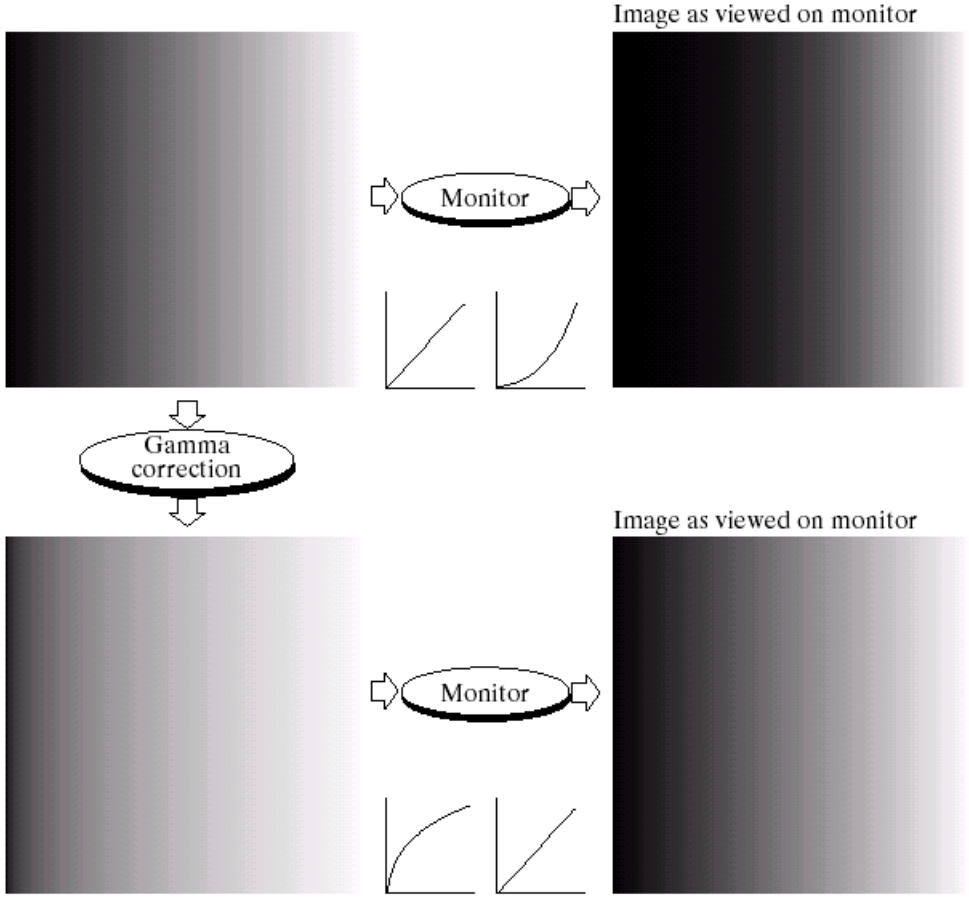

Transformed Intensities
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.2
0.4
0.6
0.8
1
Original Intensities

Gamma Correction

- Many devices used for image capture, display and printing respond according to a power law.
- The *exponent in the power-law equation is referred to as* ***gamma***.
- The process of correcting for the power-law response is referred to as **gamma correction.**
- Example: – CRT devices have an intensity-to-voltage response that is a power function (exponents typically range from 1.8-2.5).
- With reference to the $\gamma = 2.5$, the display systems would tend to display images that darker than intended. In this case Gamma correction in this case could be achieved by applying the transformation s= r ^(1/2.5) = $r^{0.4}$

Gamma Correction

- Gamma correction is important if displaying an image accurately on a computer screen is of concern.
- Images that are not corrected properly can look either bleached out, or what is more likely, too dark.
- Problem is that display devices do not respond linearly to different intensities. It can be corrected using a log transform.
- So gamma correction is required. It not only change the intensity , but also the ration of red to green to blue in color image.



Piecewise-Linear Transformation Functions

- Rather than using a well defined mathematical function we can use arbitrary user-defined transforms called piecewise-linear transformation functions which can be arbitrarily complex.
- Their specification requires considerably more user input.
- Examples:
  - Contrast stretching
  - Intensity-level slicing
  - Bit-plane slicing

Contrast Stretching

- One of the simplest piecewise linear functions.
- Low-contrast images can result from poor illumination, lack of dynamic range in the imaging sensor, or even the wrong setting of a lens aperture during image acquisition.
- It is a process that expands the range of intensity levels in an image so that it spans the full intensity range of the recording medium or display device.
- The image below show a contrast stretching linear transform to add contrast to a poor quality image.

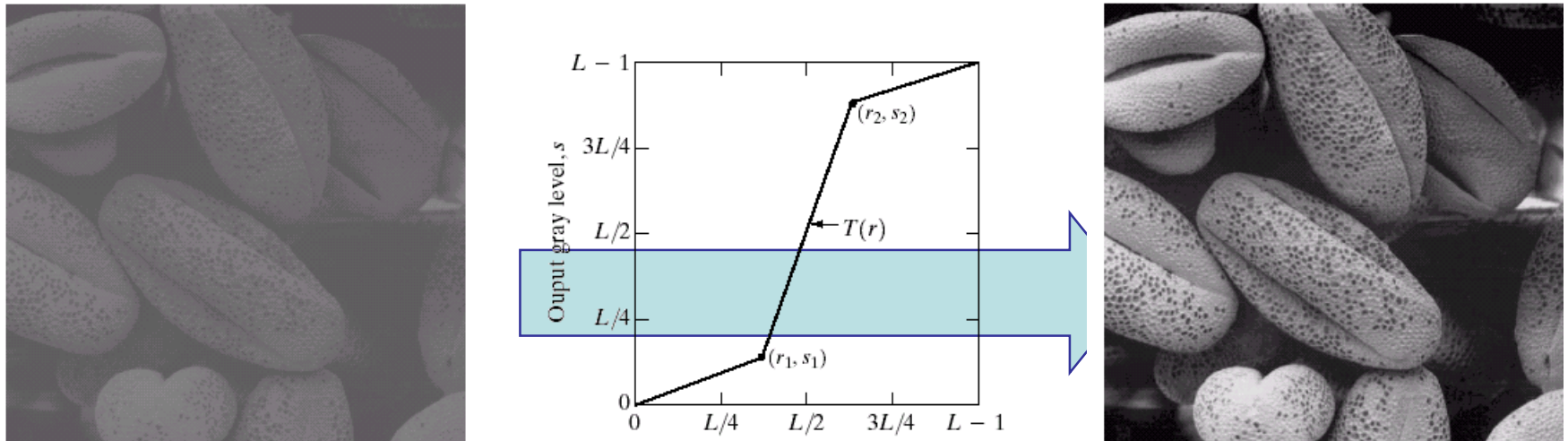


Figure: Contrast Stretching

Contrast Stretching

The control points (r_1,s_1) and (r_2,s_2) control the shape of the transform function, T(r).

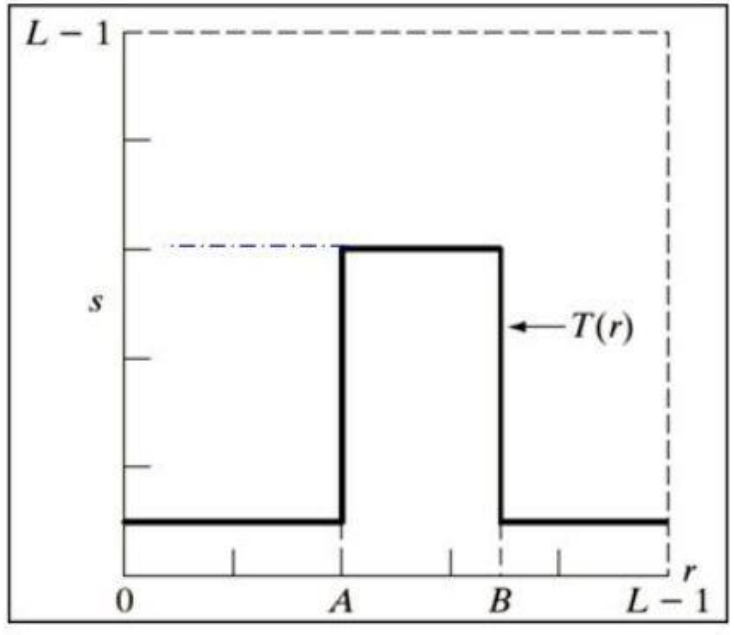
- If $r_1 = s_1$ and $r_2 = s_2$, the transformation is a linear function that produces no changes in gray levels.
- If $r_1=r_2$, $s_1=0$ and $s_2=L-1$ yields a thresholding function that creates a binary image.
- The contrast stretched image shown in the previous slide is obtained using the transformation obtained from the equation of the line having following points
  $(r_1,s_1)=(r_{min},0)$ and $(r_2, s_2)=(r_{max},L-1)$
- Intermediate values of (r_1,s_1) and (r_2,s_2) produce various degree of spread in the gray levels of the output images, thus affecting its contrast.
- In general, $r_1 \leq r_2$, and $s_1 \leq s_2$, so that function is always increasing.

Intensity/Gray Level Slicing

- It is the process of highlighting a specific range of intensities in an image that might be of interest. Examples: enhancing features such as masses of water in satellite imagery, enhancing flaws in X-rays etc.
- There are two basic approaches:
  - **The first approach** is to set all pixels values within a range of interest to one value (say, white) and all others to another value (say, black). This transformation produces a binary image.
  - **The second approach** is based on transformation in which it brighten (or darken) pixel values in a range of interest and leave all other unchanged.

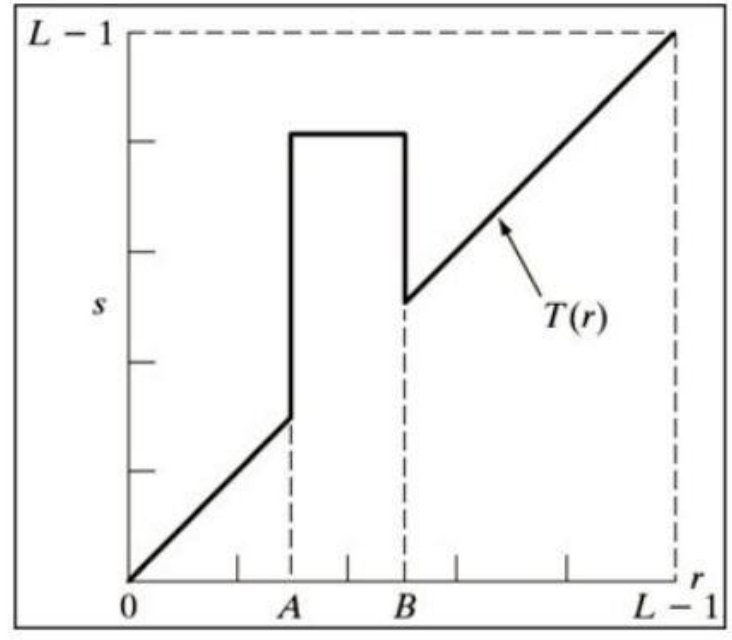
Intensity level slicing

Without Background



$$s = \begin{cases} L; A \leq r \leq B \\ 0; otherwise \end{cases}$$

With Background



$$s = \begin{cases} L; A \leq r \leq B \\ r; otherwise \end{cases}$$

Figure: (a) This intensity transformation highlights intensity range [A, B] and reduces all other intensities to a lower level. (b) This transformation highlights range [A, B] and preserves all other intensity levels.

Gray level slicing

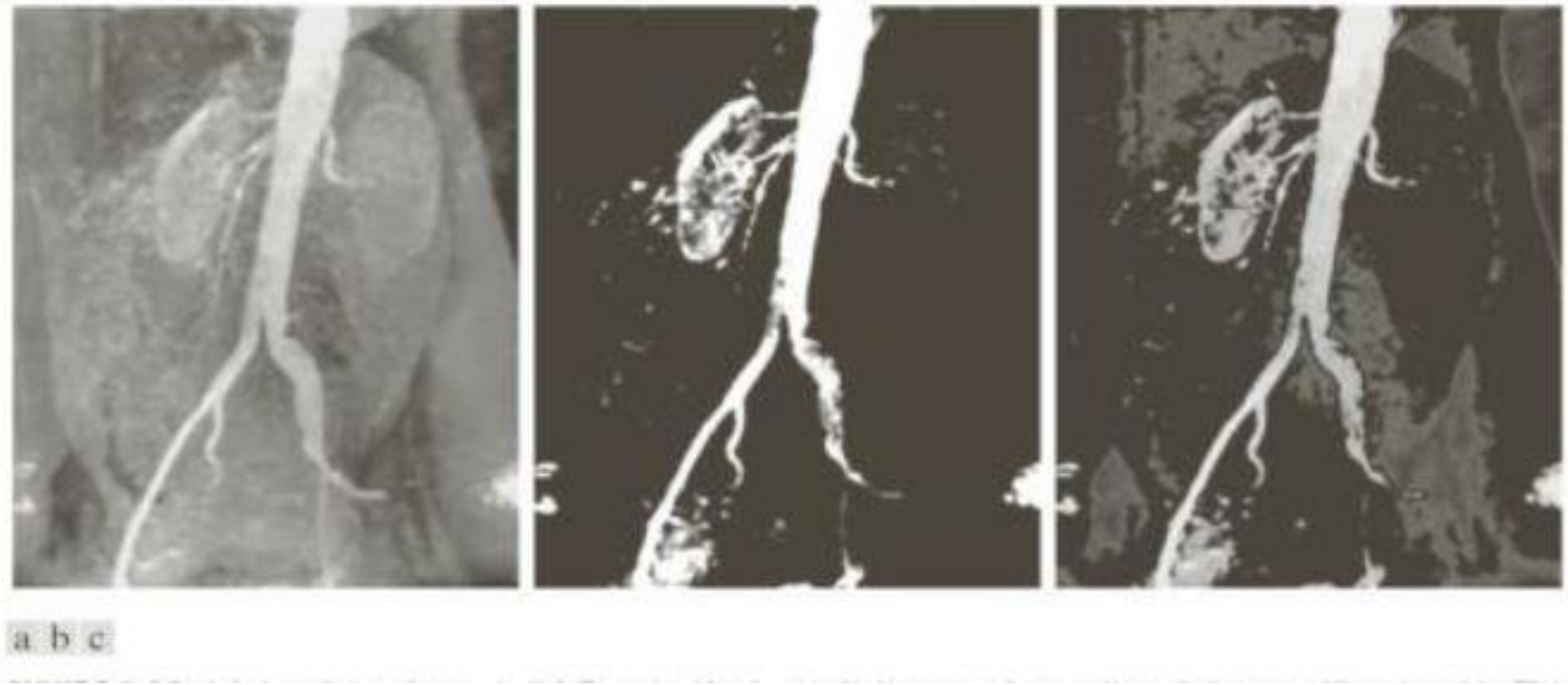


Figure: (a) Aortic Angiogram. (b) Result of using a slicing transformation of the type in fig (a) with the range of intensities of interest selected in the upper end of gray scale. (c) Result of using the transformation in fig (b), with the selected area set to black, so that grays in the area of the blood vessels and kidneys were preserved.

Bit Plane Slicing

- Pixels are digits compose of bits. For example, the intensity of each pixel in a 256-level gray-scale image is composed of 8 bits.
- Instead of highlight intensity-level ranges, we could highlight the contribution made to total image appearance by specific bits.
- For example, in the given figure, an 8-bit image may be considered as being composed of eight 1-bit planes, with plane 1 containing the lowest-order bit of all pixels in the image and plane 8 all the highest-order bits.

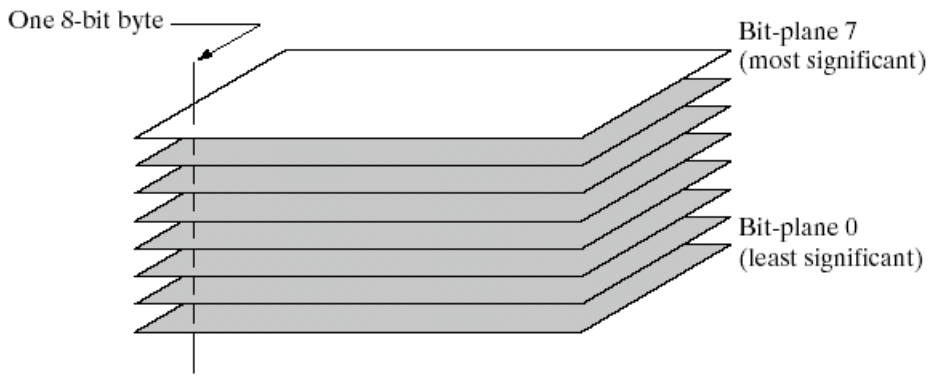
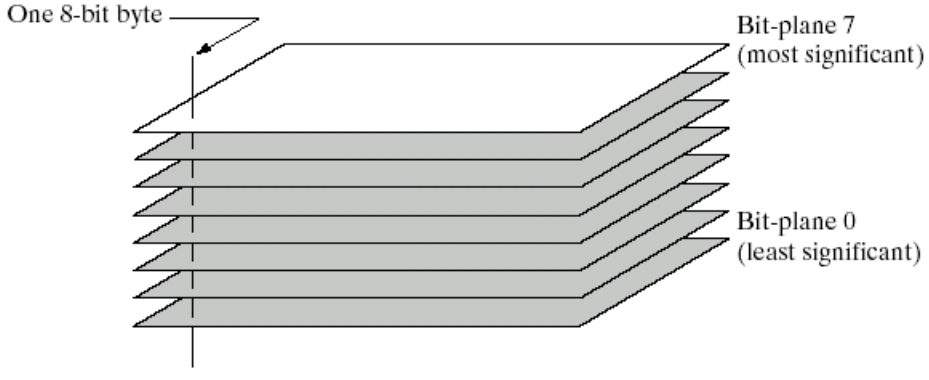


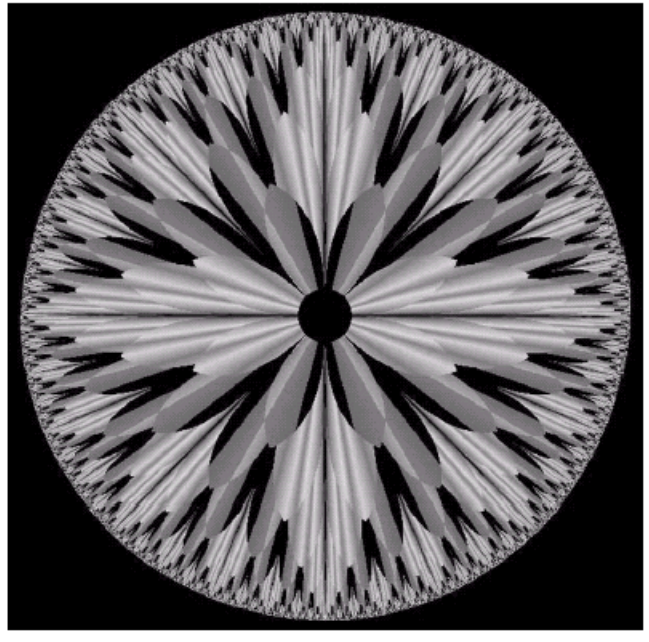
Figure: Bit-plane representation of an 8-bit image.

Bit plane slicing

Often by isolating particular bits of the pixel values in an image we can highlight interesting aspects of that image

- Higher-order bits usually contain most of the significant visual information
- Lower-order bits contain subtle details





[10000000]

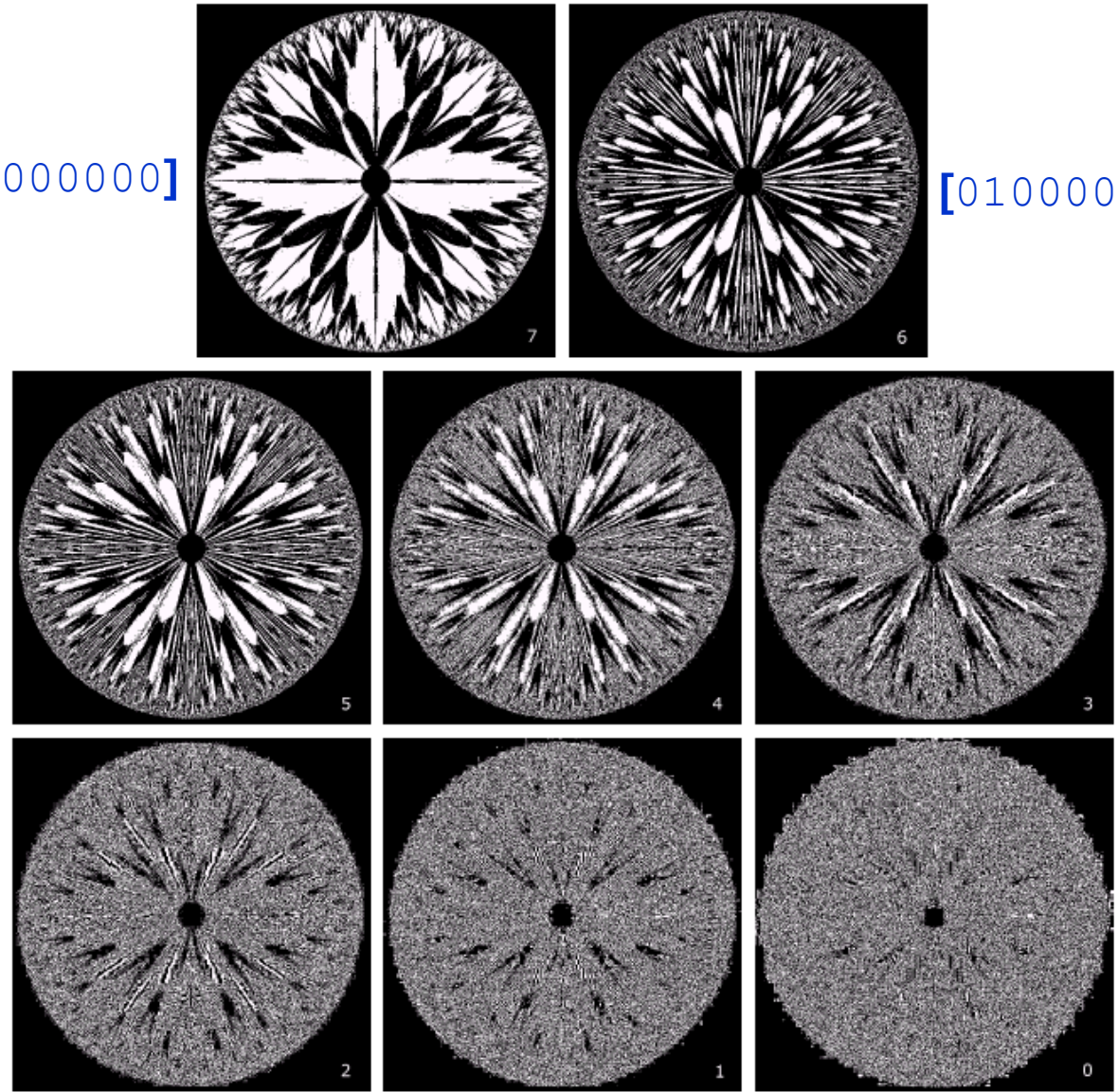
[01000000]

[00100000]

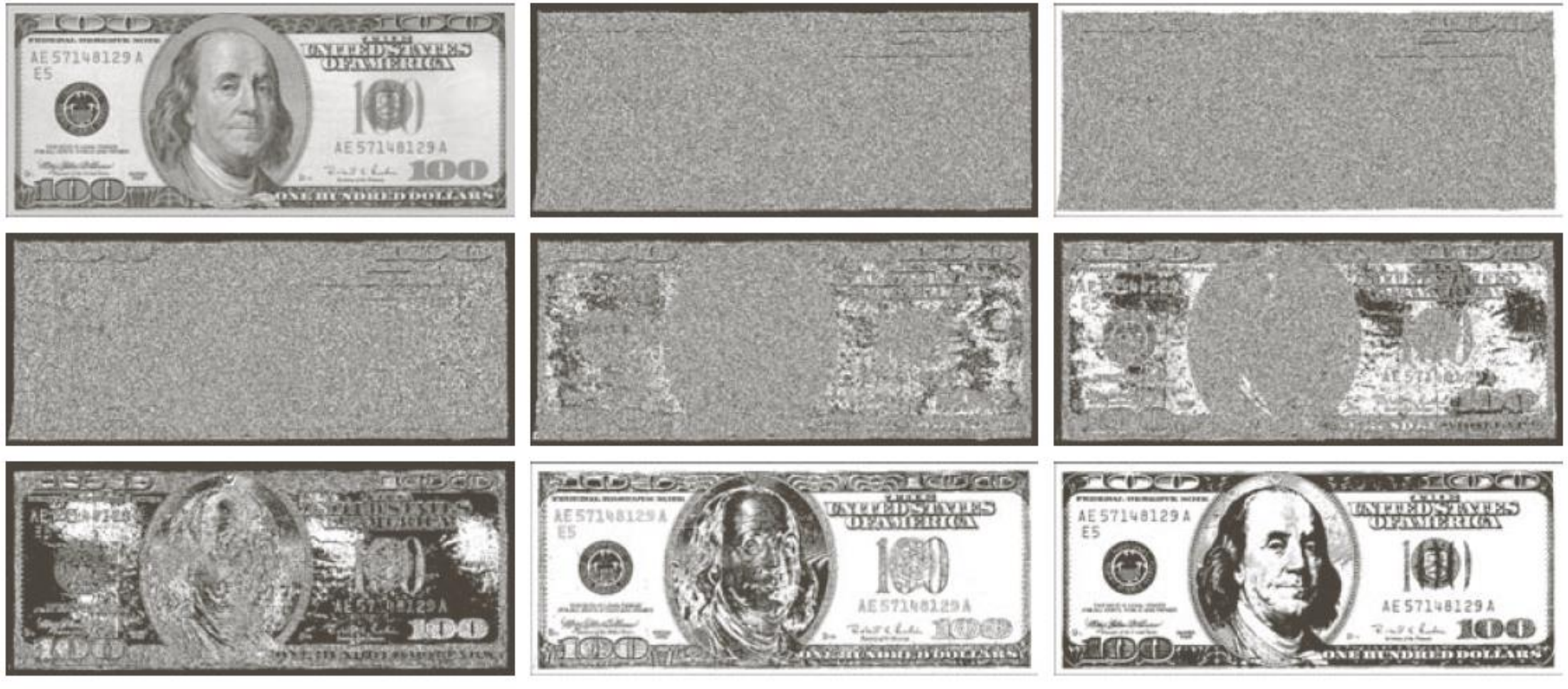
[00001000]

[00000100]

[00000001]



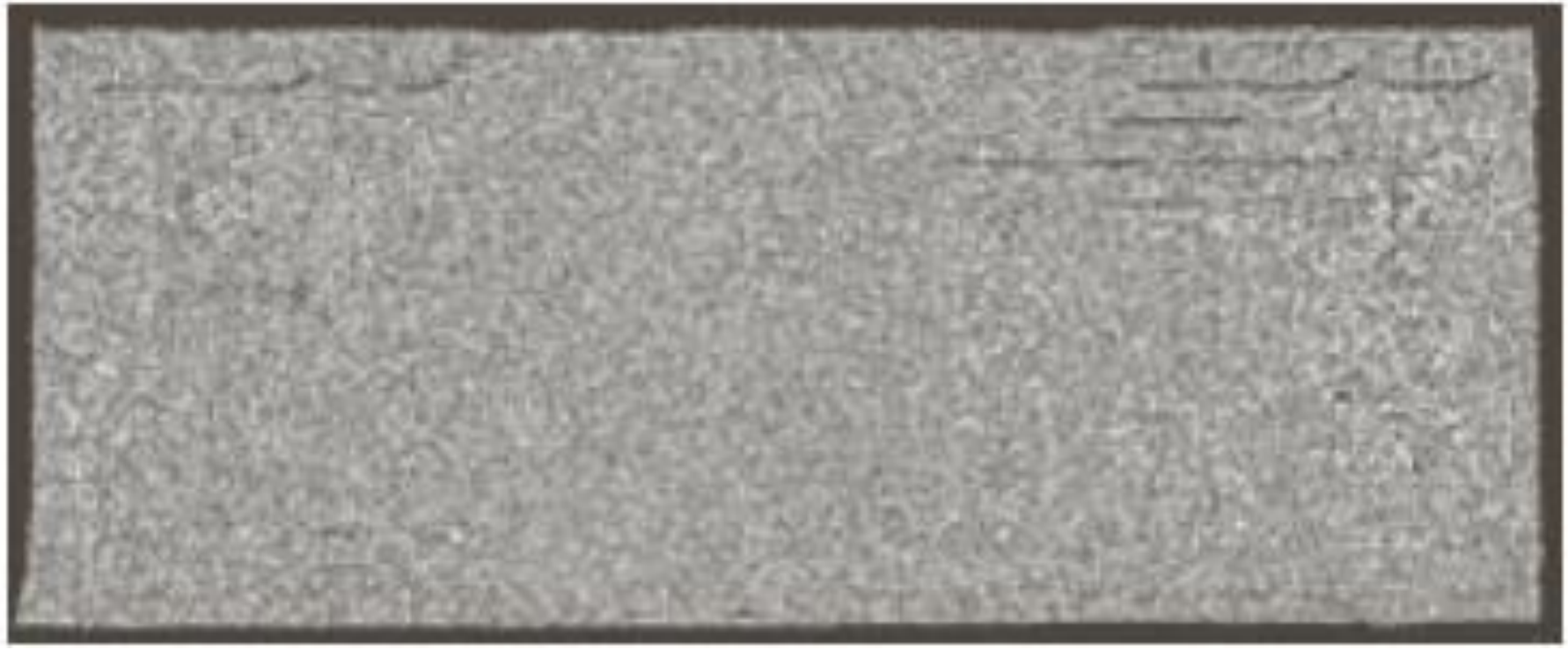
Bit Plane Slicing (cont…)



a b c
d e f
g h i

FIGURE 3.14 (a) An 8-bit gray-scale image of size 500 × 1192 pixels. (b) through (i) Bit planes 1 through 8, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image.

AE 57148129 A
E5
THE UNITED STATES OF AMERICA
100
AE 57148129 A
100
100
ONE HUNDRED DOLLARS



AE57148129A
E5
THE UNITED STATES OF AMERICA
100
AE57148129A
100
ONE HUNDRED DOLLARS

AE 57148129 A
E5
THE UNITED STATES OF AMERICA
AE 57148129 A
100
100
ONE HUNDRED DOLLARS

Reconstructed image using only bit planes 8 and 7

Reconstructed image using only bit planes 8, 7 and 6

Reconstructed image using only bit planes 7, 6 and 5

Bit Plane Slicing (contd…)

- Decomposing an image into its bit plane is useful for analyzing the relative importance of each bit in the image. This in-turn aids in determining the adequacy of the number of bits used to quantized the image.
- Also, this type of decomposition is useful for image segmentation.
- We can reconstruct an image with fewer than all planes. The reconstruction is done by multiplying by pixels of nth plane by the constant 2^{n-1}.
- For example, to obtain a reconstruct image of bit plane 8 and 7, we multiplied bit plane by 128, bit plane 7 by 64, and added the two planes.

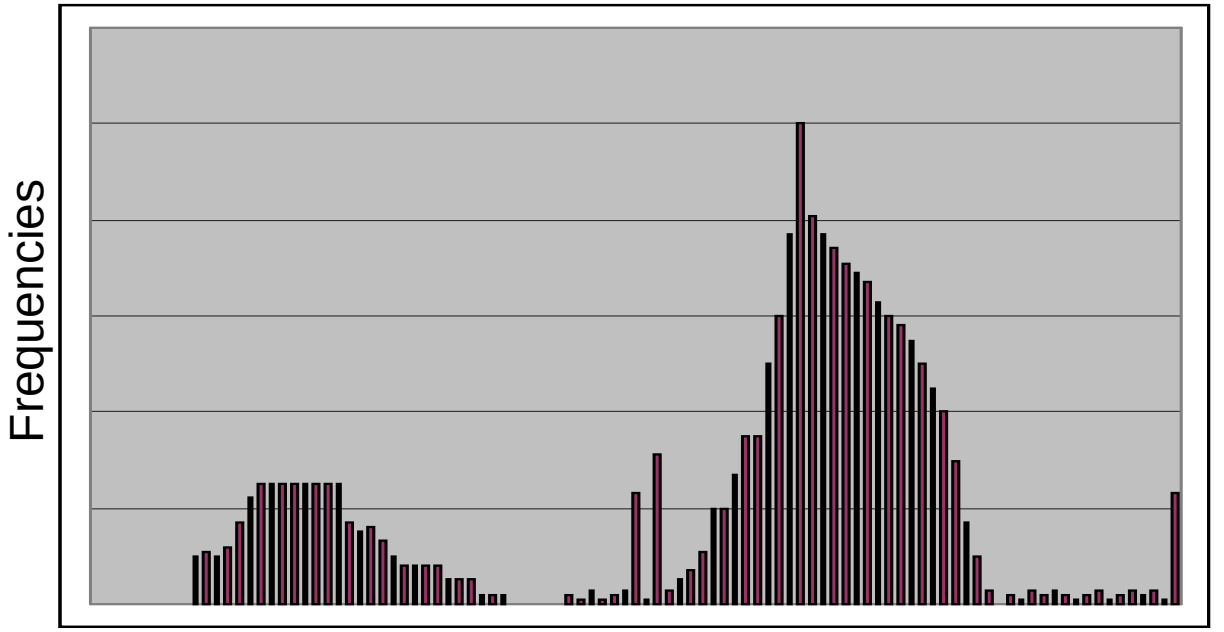
Histogram

Topic covered

- Histogram
- Histogram equalization
- Histogram modification
- Histogram Specification

Histogram

- Histogram in general are frequency distributions, and histograms of images describe the frequency distribution of the intensity values that occur in an image. It provides a global description of the appearance of an image.
- The histogram of an image represents the relative frequency of occurrence of various gray level in an image.



Histogram (contd…)

- The histogram of a digital image, f, with intensities [0, L-1] is a discrete function

  $h(r_k) = n_k$.

- Where r_k is the kth intensity value and n_k is the number of pixels in f with intensity r_k
- Normalizing the histogram is common practice i.e. it the process of dividing the each of its components by the total number of pixels in the image – Assuming an MxN image, this yields

  $p(r_k) = n_k/MN$ for $k=0,1,2,....,L-1$ – $p(r_k)$ is, basically, an estimate of the probability of occurrence of intensity level r_k in an image such that $\Sigma\, p(r_k) = 1$

Uses of Histogram Processing

- Histogram are used to depict the statistics in an easily interpreted visual format.
- With a histogram,
  - ❑it is easy to find defect in image i.e. it is easy to find if an image is properly exposed (i.e. image statistics)
  - ❑Modern digital cameras often provide a real-time histogram overlay on the viewfinder to help prevent taking poorly exposed pictures,
  - ❑It is used later to improve the visual appearance of an image (image enhancement)
  - ❑Used is image segmentation
  - ❑Used in image compression

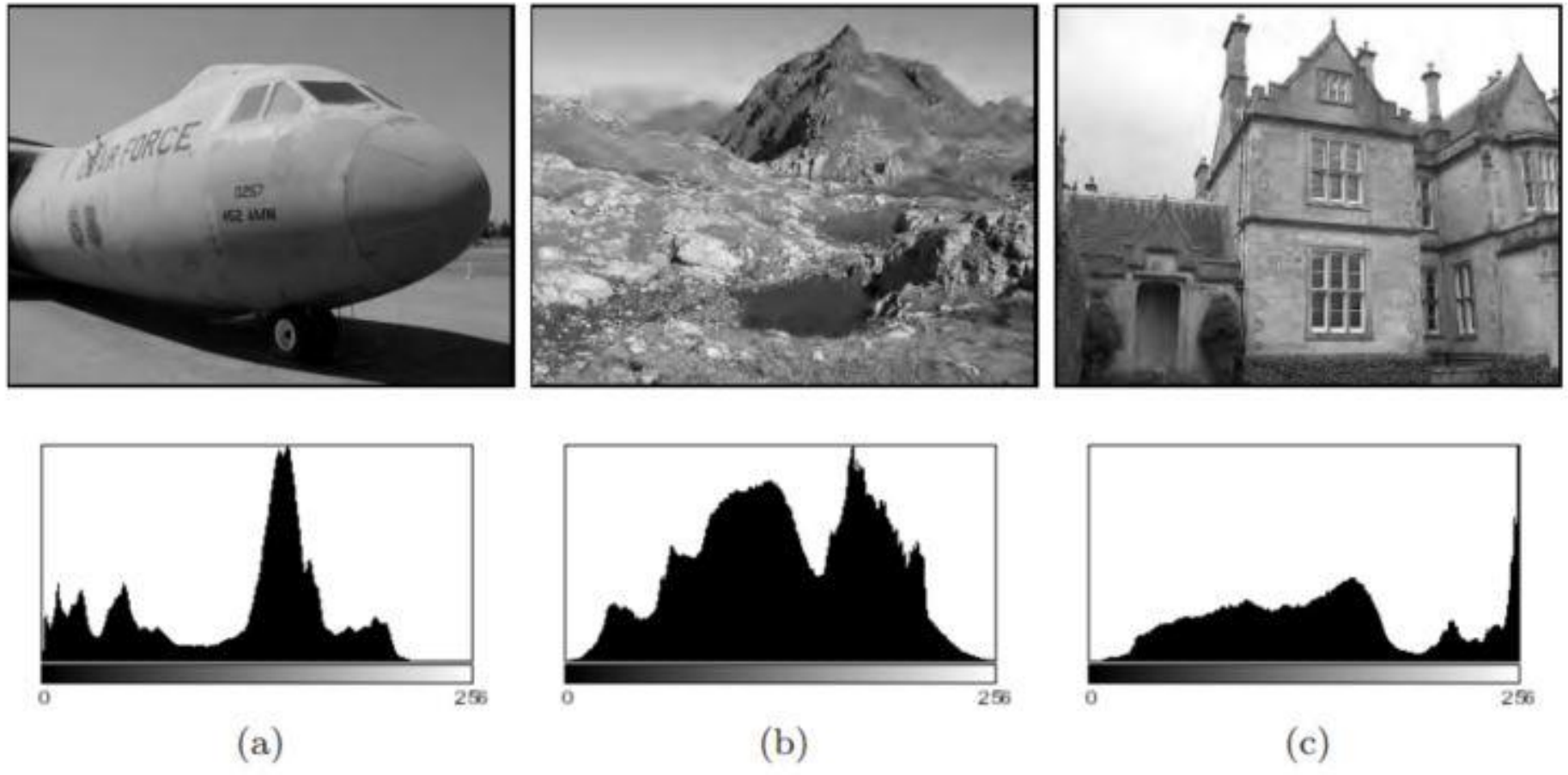
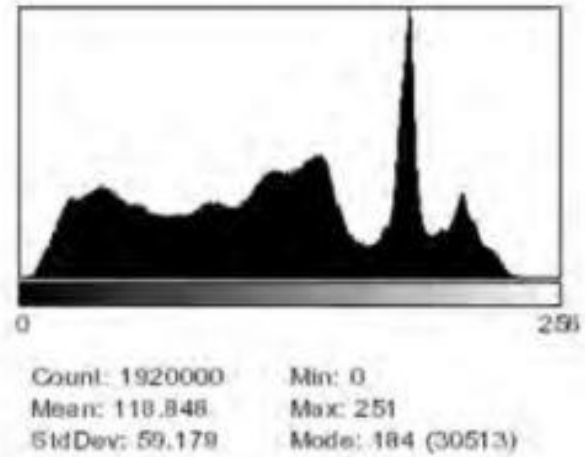


Figure: Exposure errors are readily apparent in histogram. (a) Underexposed photograph, (b) Properly Exposed Photograph, (c) Overexposed photograph

(a)



(b)

Figure: (a) Digital Camera back display showing a histogram overlay, (b) An 8 –bit gray-scale image and a histogram depicting the frequency distribution of its 256 intensity values.

Plotting Histogram

<u>The histogram of an image can be plotted as</u>:

1. The X-axis represents the gray level and Y-axis has the number of pixel in each gray level i.e. $\boldsymbol{h(r_k) = n_k}$ **versus** $\boldsymbol{r_k}$
2. The X-axis represents the gray level and Y-axis represents the probability of occurrence of that gray level i.e. $\boldsymbol{p(r_k) = n_k / MN}$ **versus** $\boldsymbol{r_k}$

Example of histogram plot

- Suppose an image has:

Gray Level (r_k)	No. of pixels (n_k)
0	40
1	20
2	10
3	15
4	10
5	3
6	2

Method -1

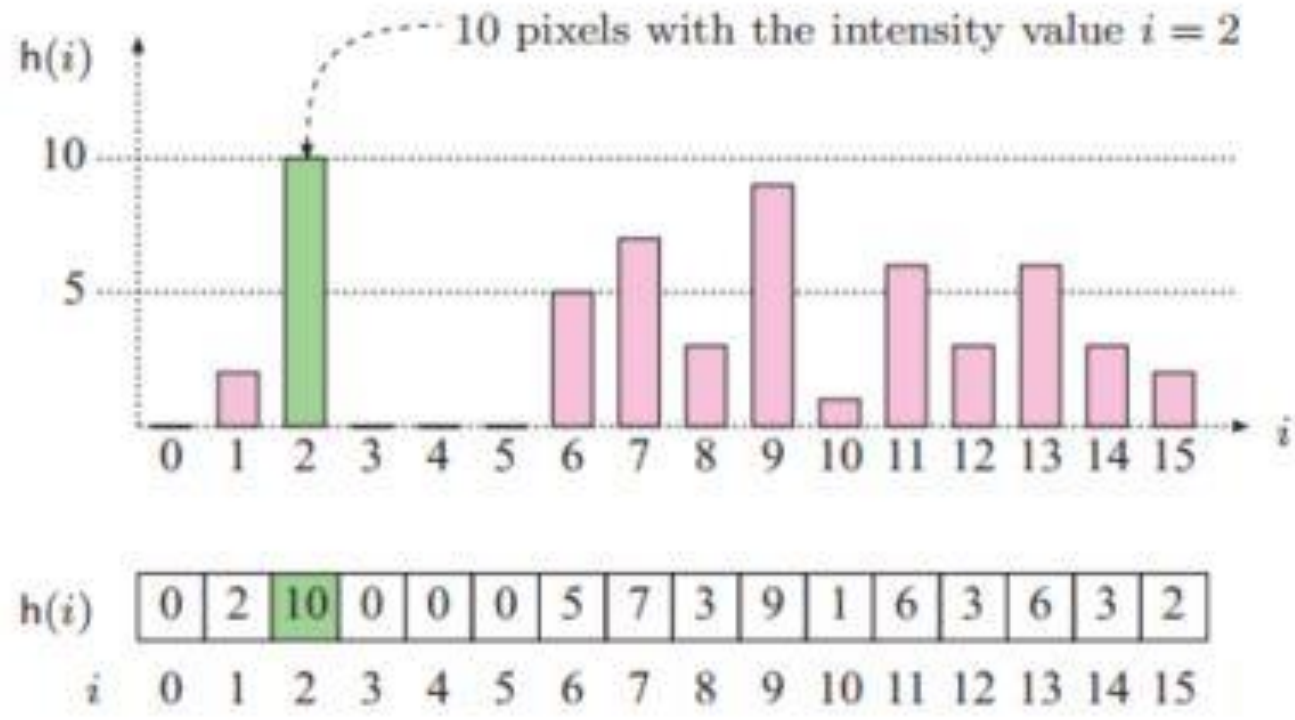
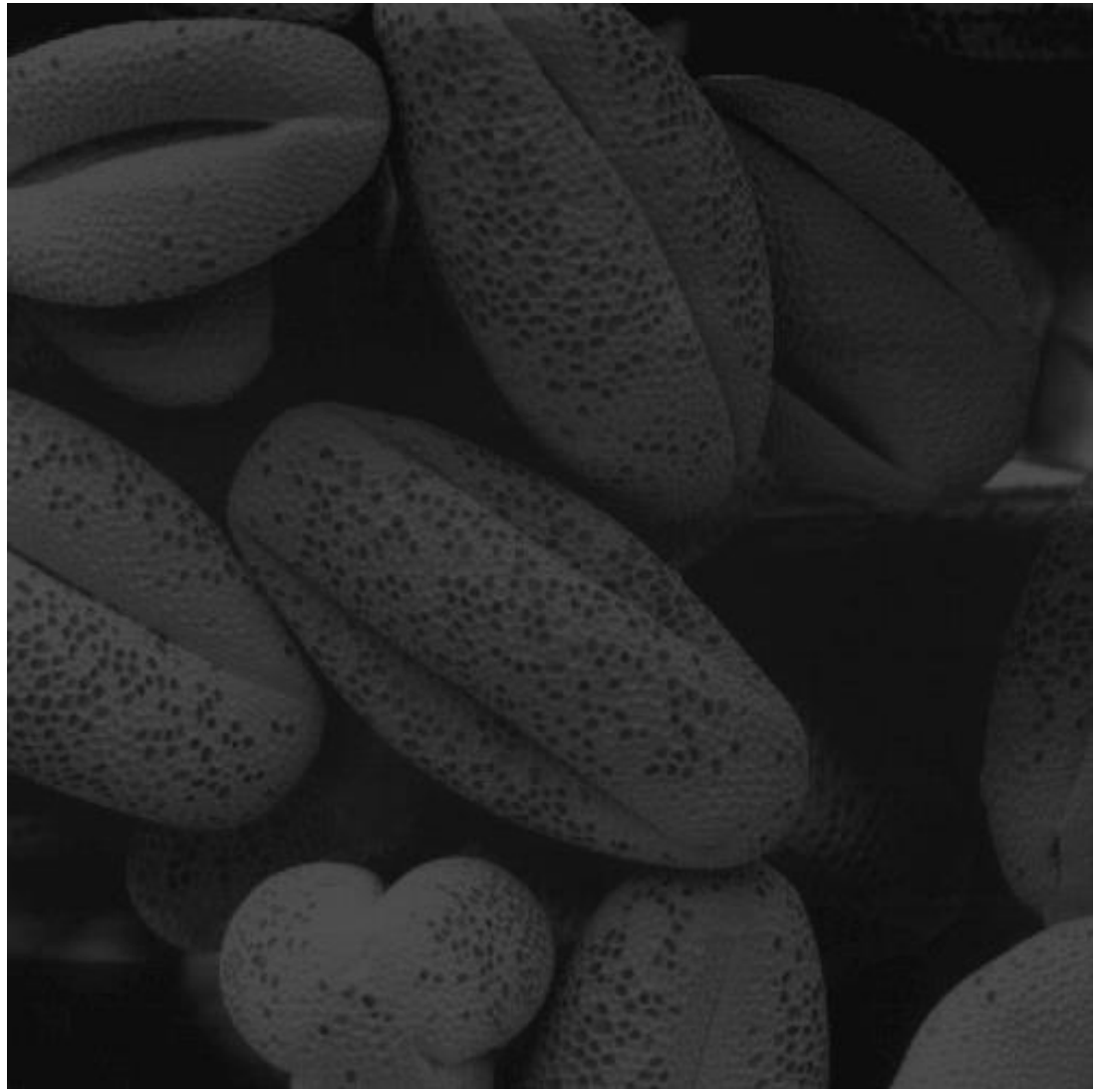


Figure: Histogram vector for an image with K = 16 possible values. The indices of the vector element i = 0, 1, ….., 14, 15 represents intensity values. The value of 10 at index 2 means that the image contains 10 pixels of intensity value 2.

Histogram Examples



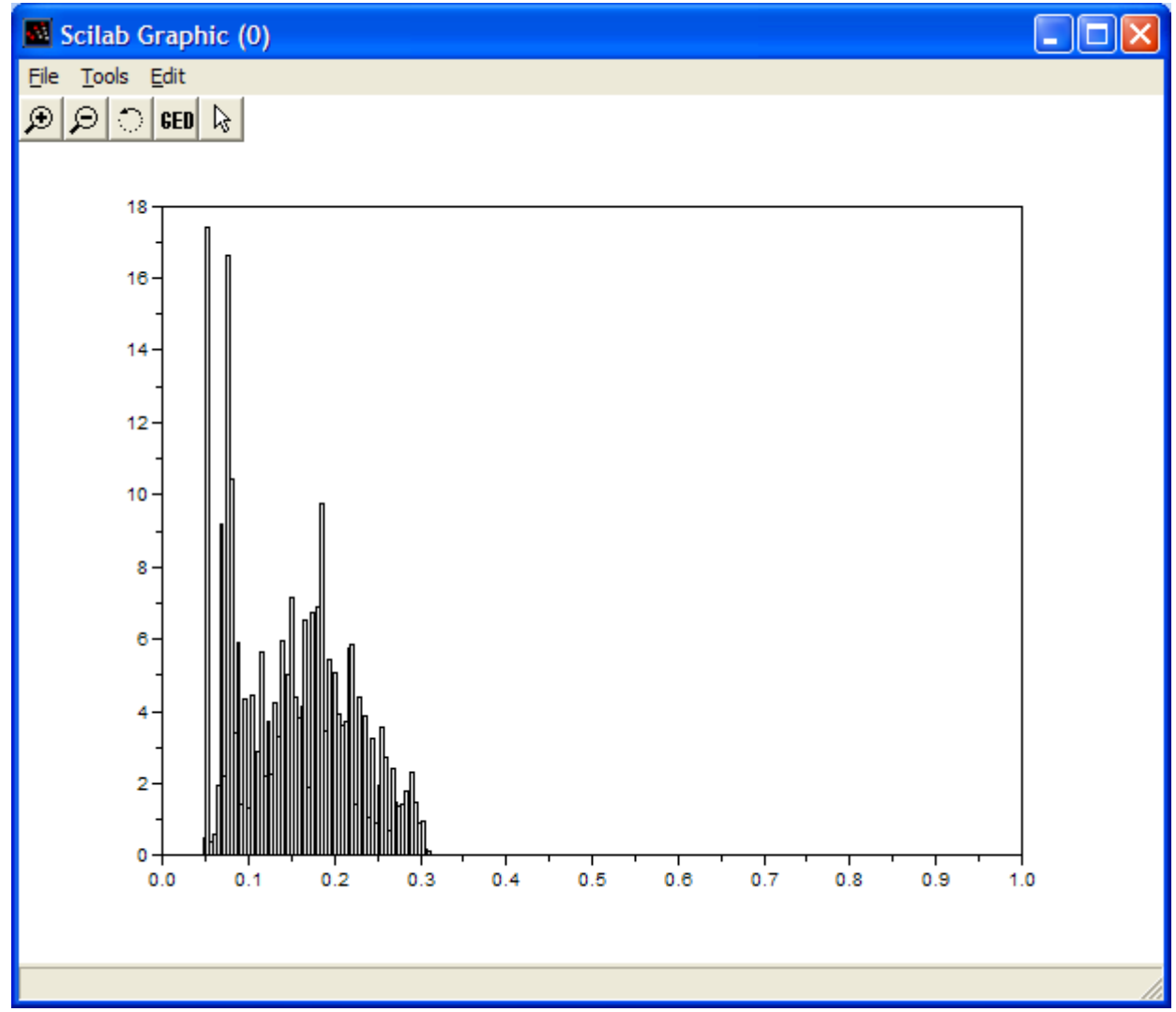


Figure: Histogram of dark image

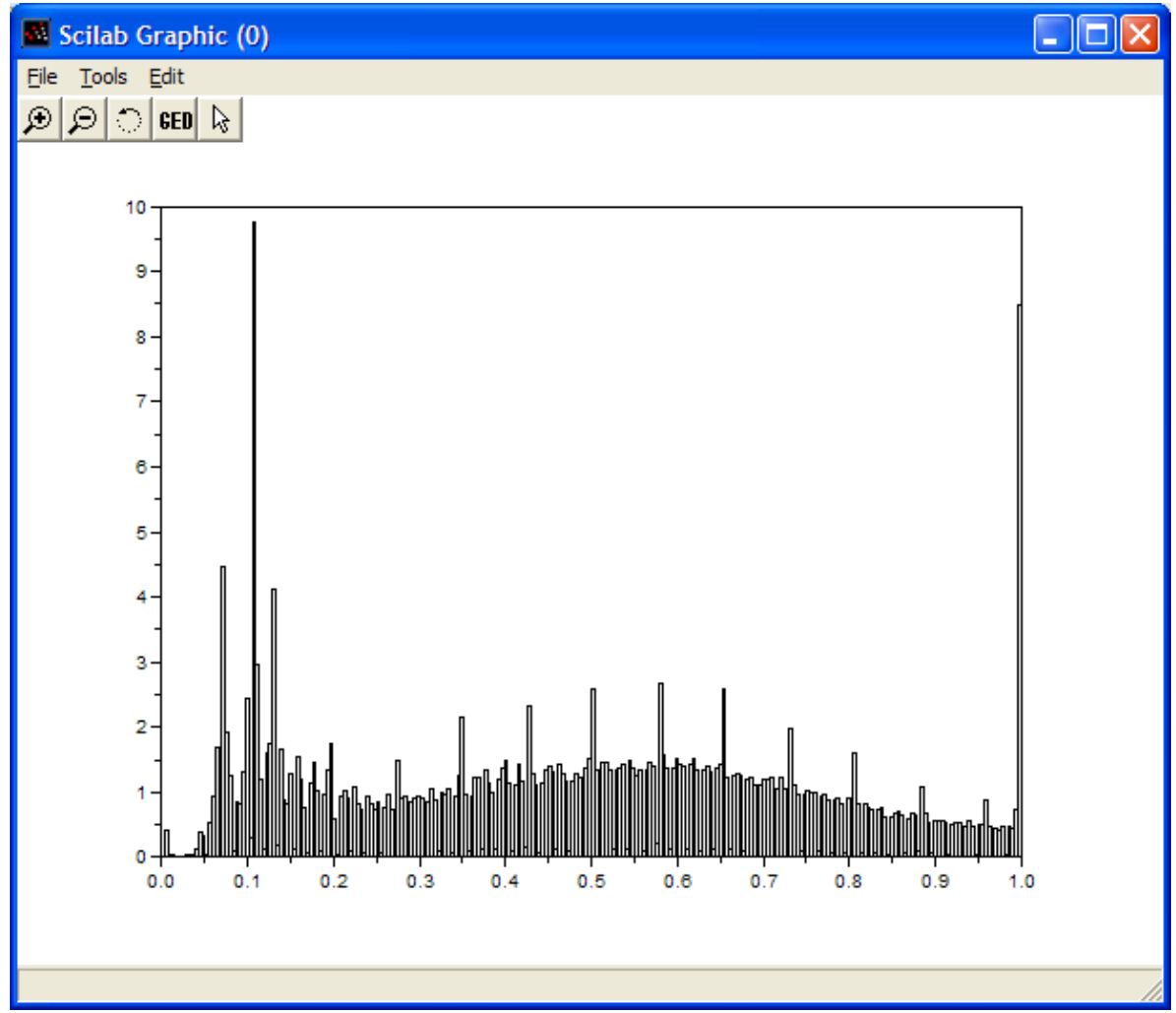


Figure: Histogram of high-contrast image

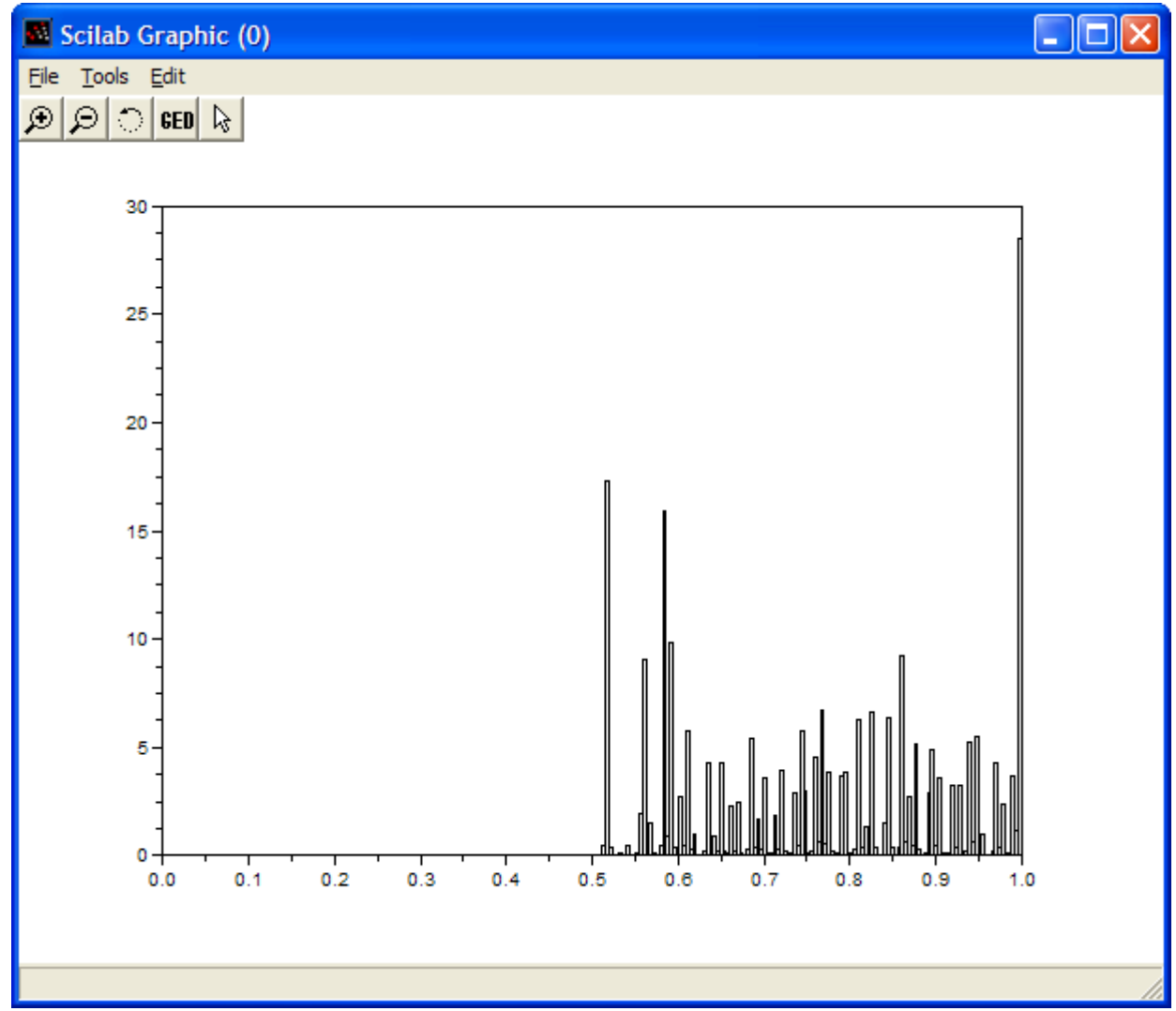
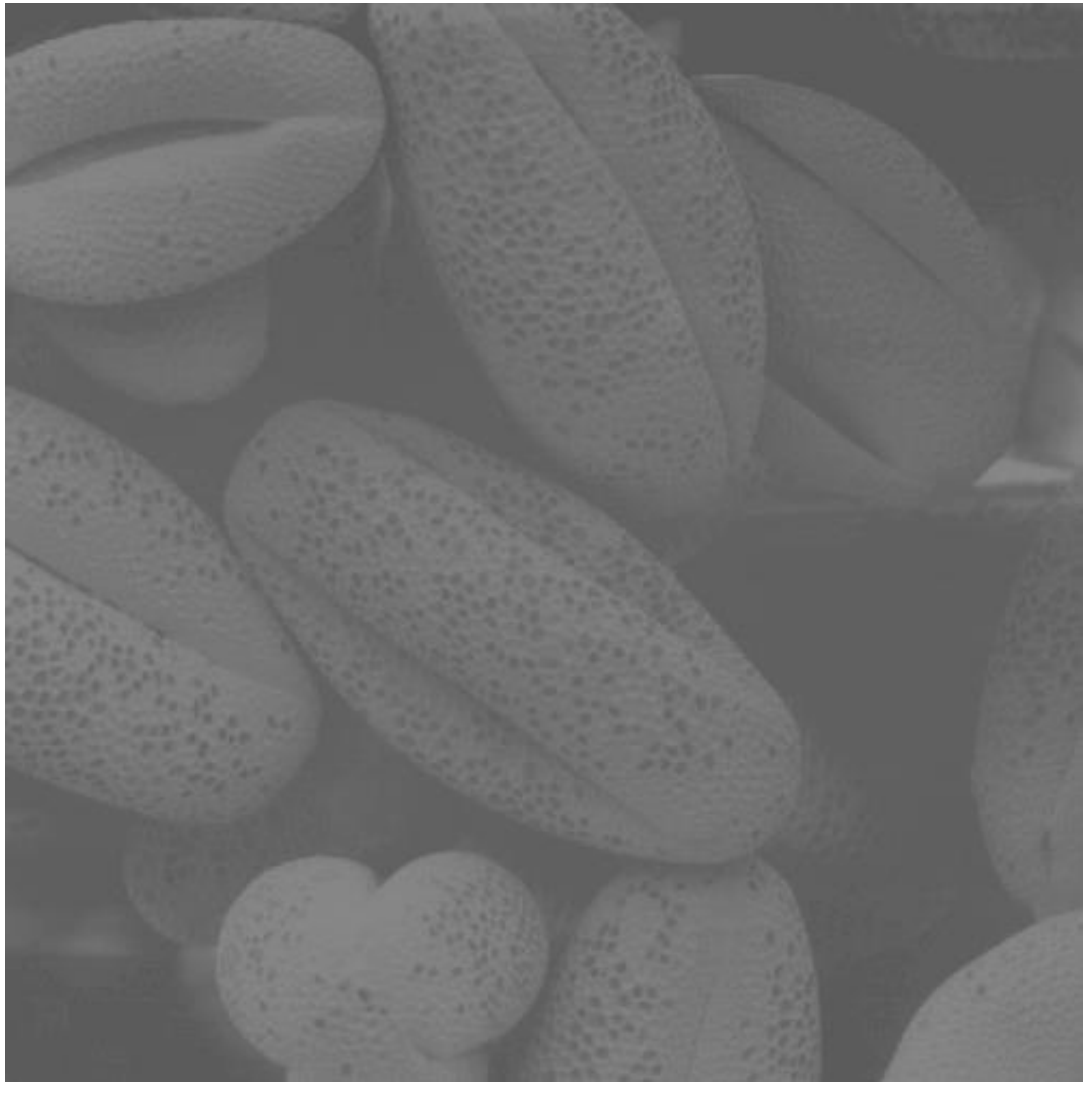


Figure: Histogram of light image



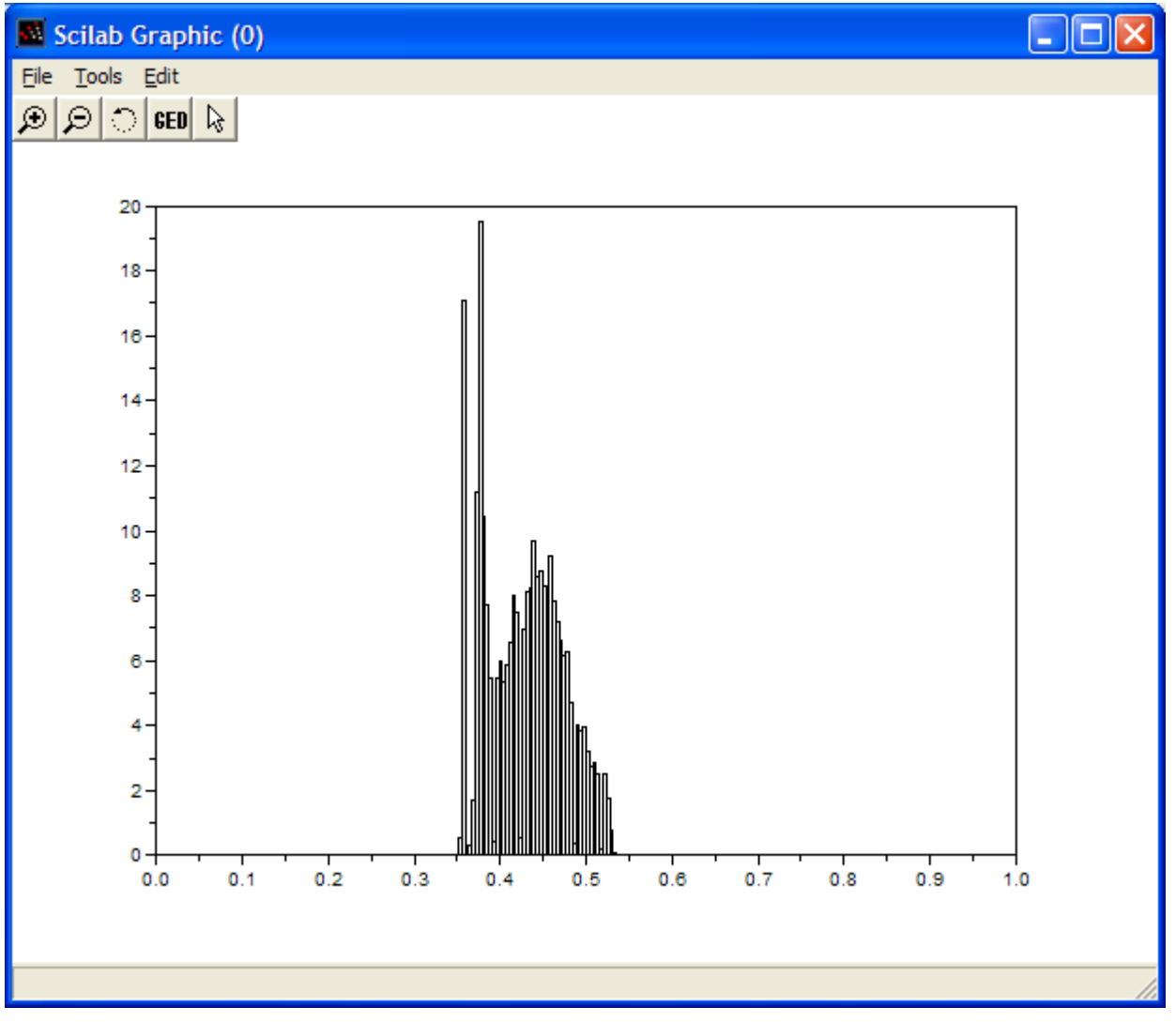


Figure: Histogram of low-contrast image

Histogram Techniques

- Histogram stretching / Linear Stretching
- Histogram Equalization / Histogram Linearization
- Histogram Specification / Histogram Matching

Contrast Stretching

<u>Contrast</u>

- Contrast is the range of intensity values *effectively* used within a given image, that is, the difference between the image's maximum and minimum pixel values.
- A full-contrast image makes effective use of the entire range of available intensity values from $a = a_{min}$ …… $a_{max} = 0$……. K – 1 (black to white).

Dynamic Range

- The dynamic range of an image is, in principle, understood as the number of *distinct* pixel values in the image.
- In the ideal case, the dynamic range encompasses all K usable pixels values, in which case the value is completely utilized.
- While the contrast of an image can be increased by transforming its existing values to that they utilize more of the underlying value range available, the dynamic range of image can only be increased by introducing artificial values using methods such as interpolation.
- An image with a high dynamic range is desirable because it will suffer less image-quality during image processing and compression.

Histogram Stretching / Linear Stretching

- Histogram stretching is a point of operation whose task is to modify the pixels such that the available range of values is fully covered. It is the process of increasing the dynamic range of an image.
- In this method, we do not change the basic shape of the histogram but spread it over the entire dynamic range.
- This is done by mapping current darkest and brightest pixels to the lowest and highest pixels available intensity values, respectively, and linearly distributing the intermediate values.

Histogram Stretching / Linear Stretching

- For this, we use this technique on the basis of slope. It can be gives as:

$$\text{Slope (m)} = \frac{S_{max} - Smin}{r_{max} - rmin}$$

- For transformation of histogram stretching then, transformation can be defined as:

$$S = T(r) = m\,(r - r_{min}) + S_{min}$$

Where,

S_{max} = Maximum gray level in output image

S_{min} = Minimum gray level in output image

r_{max} = Maximum gray level in input image

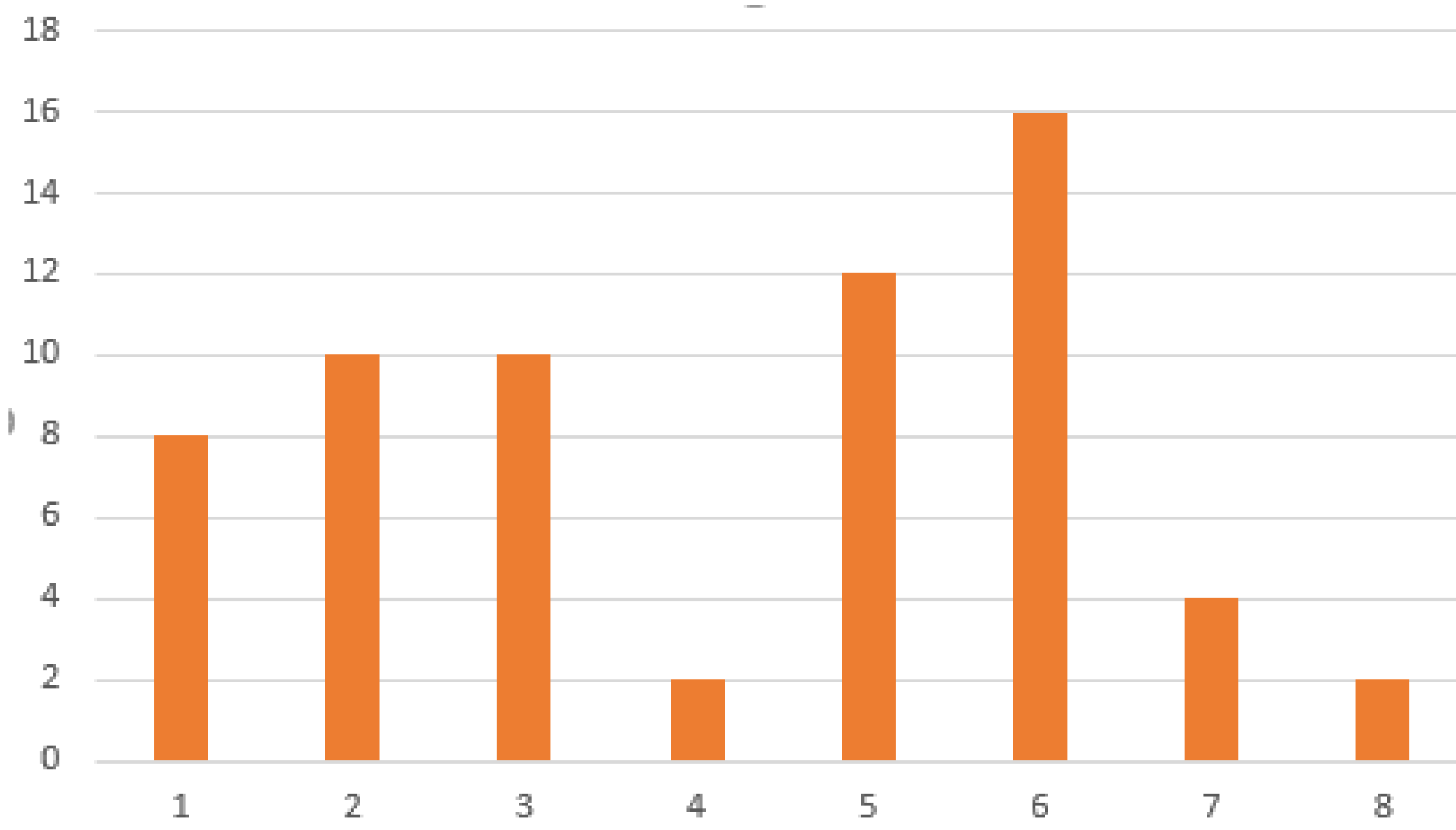
r_{min} = Minimum gray level in input image

- This transformation function stretches or increases the dynamic range of given image.

Numerical Example

Gray Level	0	1	2	3	4	5	6	7
No. of Pixels	8	10	10	2	12	16	4	2

Now, perform histogram stretching. Show that the new image has a dynamic range of entire range i.e. [0, 7] . Show the plot.


18
16
14
12
10
8
6
4
2
0
1
2
3
4
5
6
7
8

Histogram Equalization

Background

- The intensity transformation function we are constructing is of the form

$$s=T(r) \qquad 0 \le r \le L-1 \ .$$

that produce an output intensity level s for every pixel in the input image having intensity r.

- We assume that:

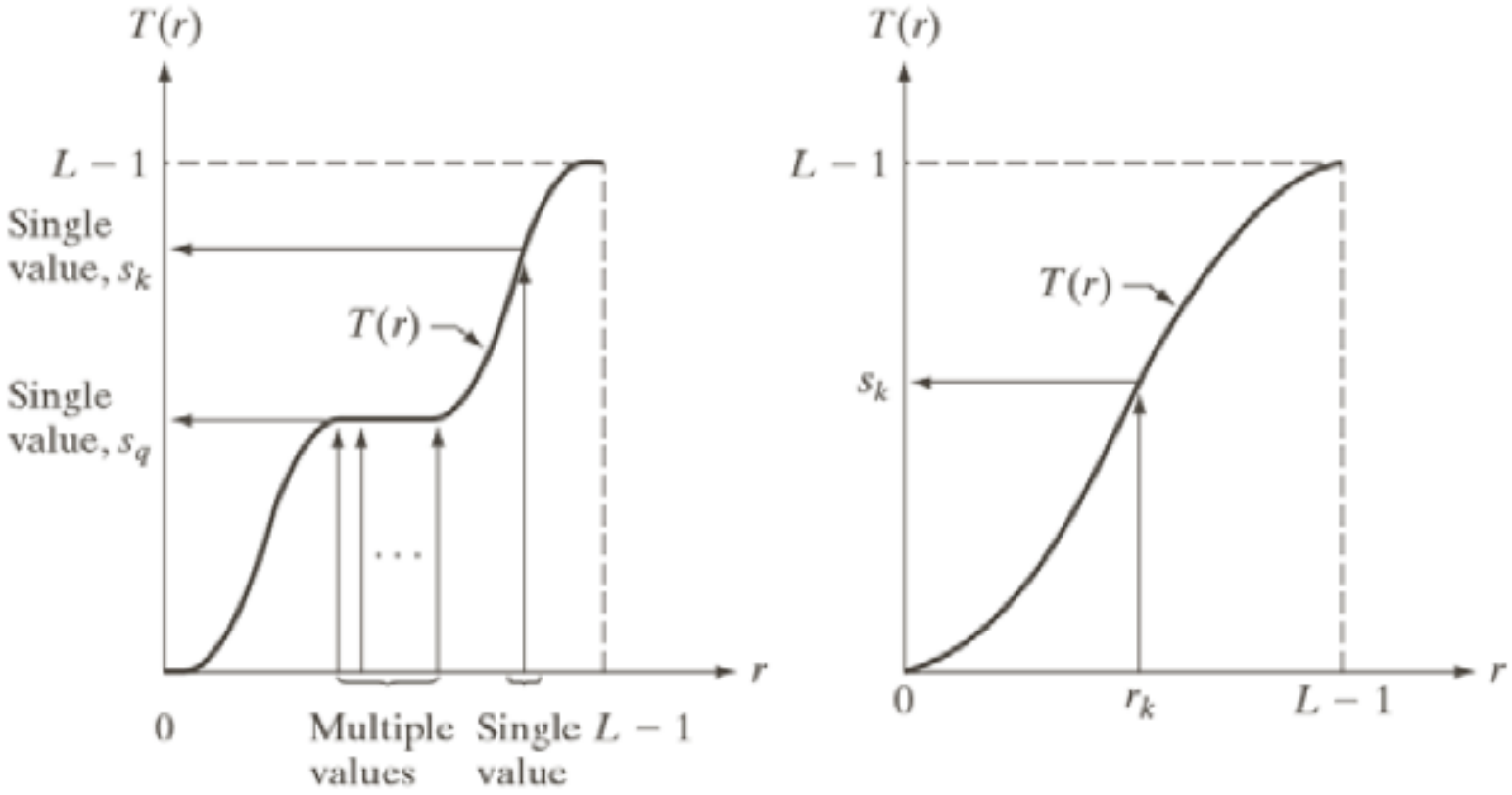
a. T(r) is monotonically increasing in the interval $0 \le r \le L\text{-}1$, and

b. $0 \le \mathrm{T}(r) \le L\text{-}1$ for $0 \le r \le L\text{-}1$

If we define the inverse $r = T^{-1}(s) \qquad 0 \le s \le L-1$ then T(r) should monotonically increasing.

- The requirement in (a) that T(r) be single valued is needed to guarantee that the inverse transformation will exist, and the monotonicity condition preserves the increasing order from black to white in the output image.
- Condition (b) guarantees that the output gray levels will be in the same range as the input levels.
- The inverse $r = T^{-1}(s) \qquad 0 \le s \le L-1$ guarantees that the mappings from s to r back to back will be one-to-one, thus preventing ambiguities.

Histogram Equalization (contd…)



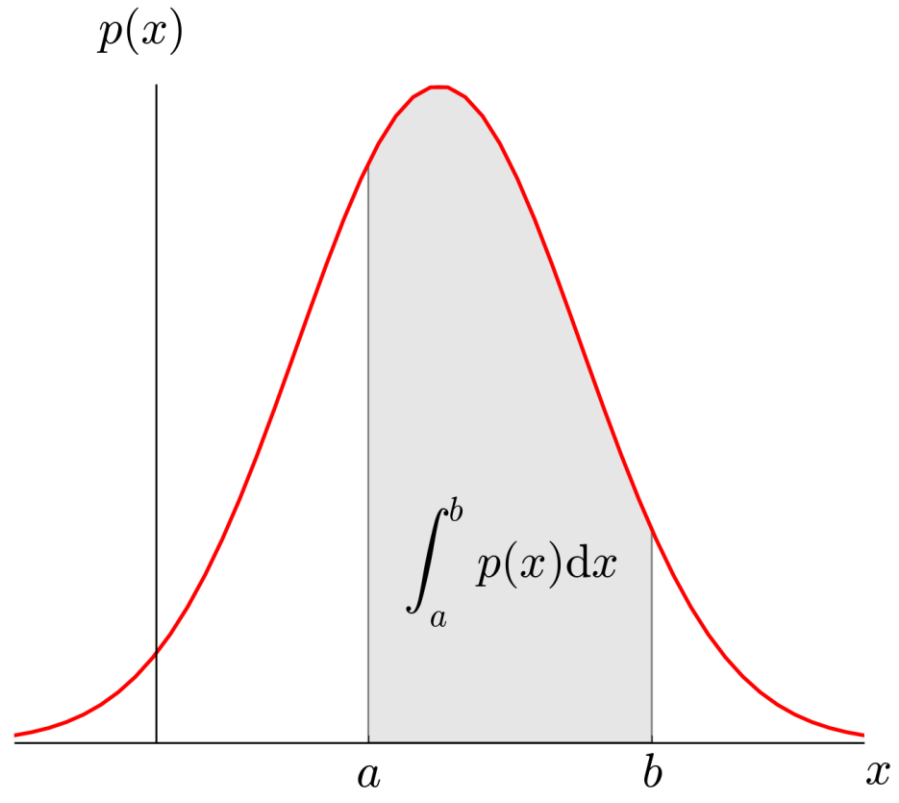
a b

FIGURE 3.17 (a) Monotonically increasing function, showing how multiple values can map to a single value. (b) Strictly monotonically increasing function. This is a one-to-one mapping, both ways.

Histogram Equalization (contd…)

<u>Probability Density Function (PDF)</u>

- Let X be a continuous random variable. Then probability distribution or probability density function (pdf) of X is a function f(x) such that for any two numbers a and b with $a \leq b$,
- $P(a \leq X \leq b) = \int_a^b f(x)dx$
- That is the probability X takes on a value in the interval [a, b] is the area above this interval and under the graph of the density function. The graph of f(x) referred to as the density curve.



Let,

$p_r(r)$ denote the probability density function of the variable r.

$p_s(s)$ denote the probability density function of the variable s.

- If $p_r(r)$ and $T(r)$ is known then $p_s(s)$ can be obtained as:

$$p_s(s) = p_r(r) \left|\frac{dr}{ds}\right| \text{ ..... (a)}$$

- A transformation function of particular importance in image processing has the form

$$s = T(r) = \text{L-1}\int_0^r p_r(w)dw \text{ , } 0 \le r \le 1, \text{ ..... Eq. 1}$$

where w is a dummy variable of integration.

Eq. 1 has the following properties:

i. $0 \le r \le 1$

ii. $r_2 \ge r_1 \rightarrow T(r_2) \ge T(r_1)$, i.e. the function T(r) is increasing with r.

iii. $s = T(0) = \int_0^0 p_r(w)dw = 0$ and $S = T(1) = \int_0^1 p_r(w)dw = 1$.

Moreover, if the original image has intensities only with in a certain range [r_{min}, r_{max}] then $s = T(r_{min}) = \int_0^{r_{min}} p_r(w)dw = 0$ and $s = T(r_{max}) = \int_0^{r_{max}} p_r(w)dw = 1$. therefore, the new intensity s takes always all values within the available range [0, 1]

- Here, using Leibniz's rule,

$$\frac{ds}{dr} = \frac{dT(r)}{dr} = [L\text{-}1]\,\frac{d}{ds}\left[\int_0^r p_r(w)dw\right] = [L\text{-}1]\,p_r(r)$$

Substituting the value of $\frac{dr}{ds}$ in equation (a), we get,

$$p_s(s) = p_r(r)\ \left|\frac{dr}{ds}\right| = p_s(s) = p_r(r)\ \left|\frac{1}{[L-1]\,pr(r)}\right| = \frac{1}{L-1} \qquad 0 \le s \le L\text{-}1$$

is a uniform probability density function.

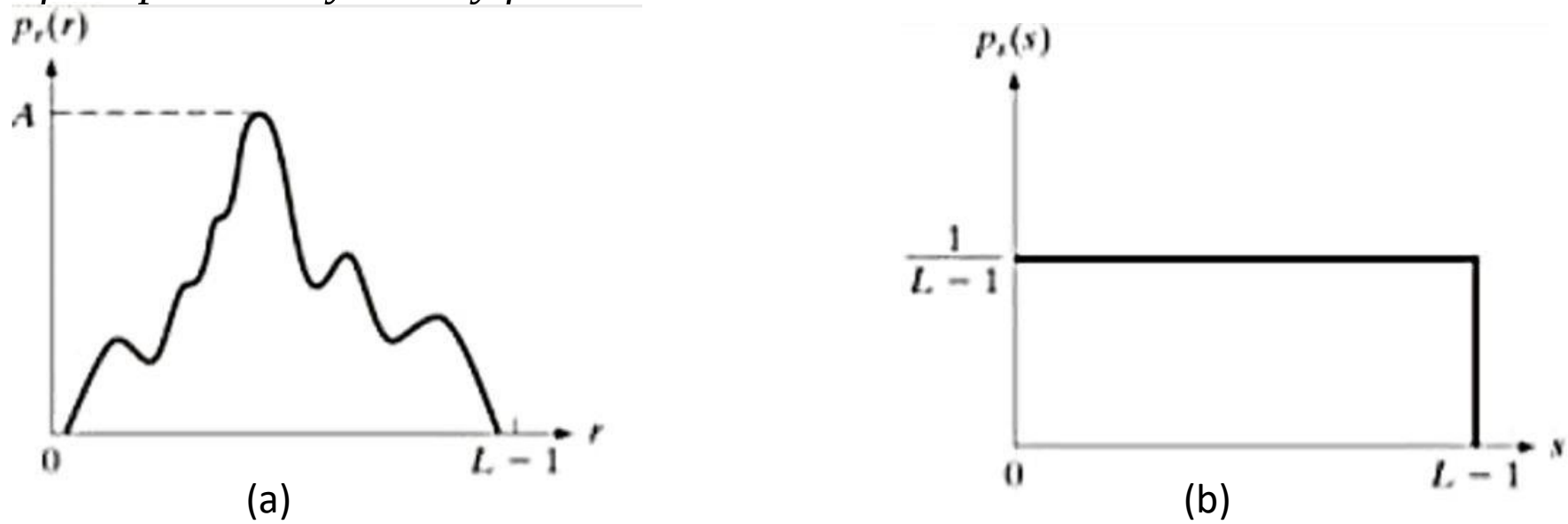


Figure: (a) An arbitrary PDF, (b) Result of applying the transformation in eq 1 to all intensity levels, r. The resulting intensities, *s*, have a uniform PDF, independently of the form of the PDF of the *r's*

Histogram Equalization (contd…)

- For discrete values, we deal with probabilities (histogram values) and summations instead of probability density functions and integrals. As mentioned earlier, the probability of occurrence of intensity level r_k in a digital image is approximated by

$$p_r(r_k) = \frac{n_k}{MN} \qquad k = 0, 1, 2, \ldots\ldots, L-1$$

- Where, MN is the total number of pixels in the image, n_k is the number of pixels that have intensity r_k and L is the number of possible intensity levels in the image.
- Histogram equalization requires construction of a transformation function s_k as
- $s_k = T(r_k) = (L-1) \sum_{j=0}^{k} p_r(r_j)$

$= (L-1) \sum_{j=0}^{k} n_j / MN$

$= \frac{(L-1)}{MN} \sum_{j=0}^{k} n_j$

- **The transformation (mapping) $T(r_k)$ in this equation is called a histogram equalization or histogram linearization.**

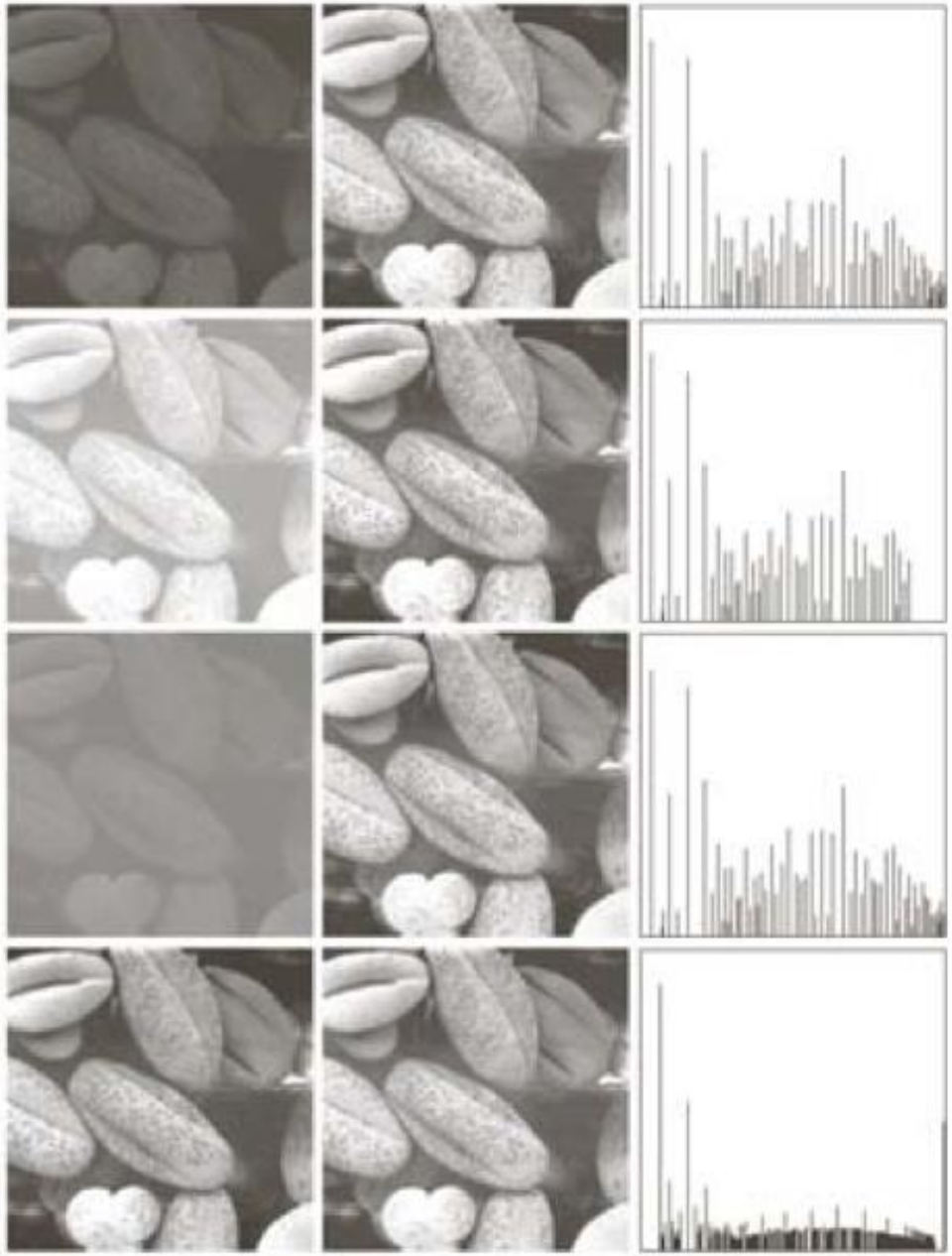
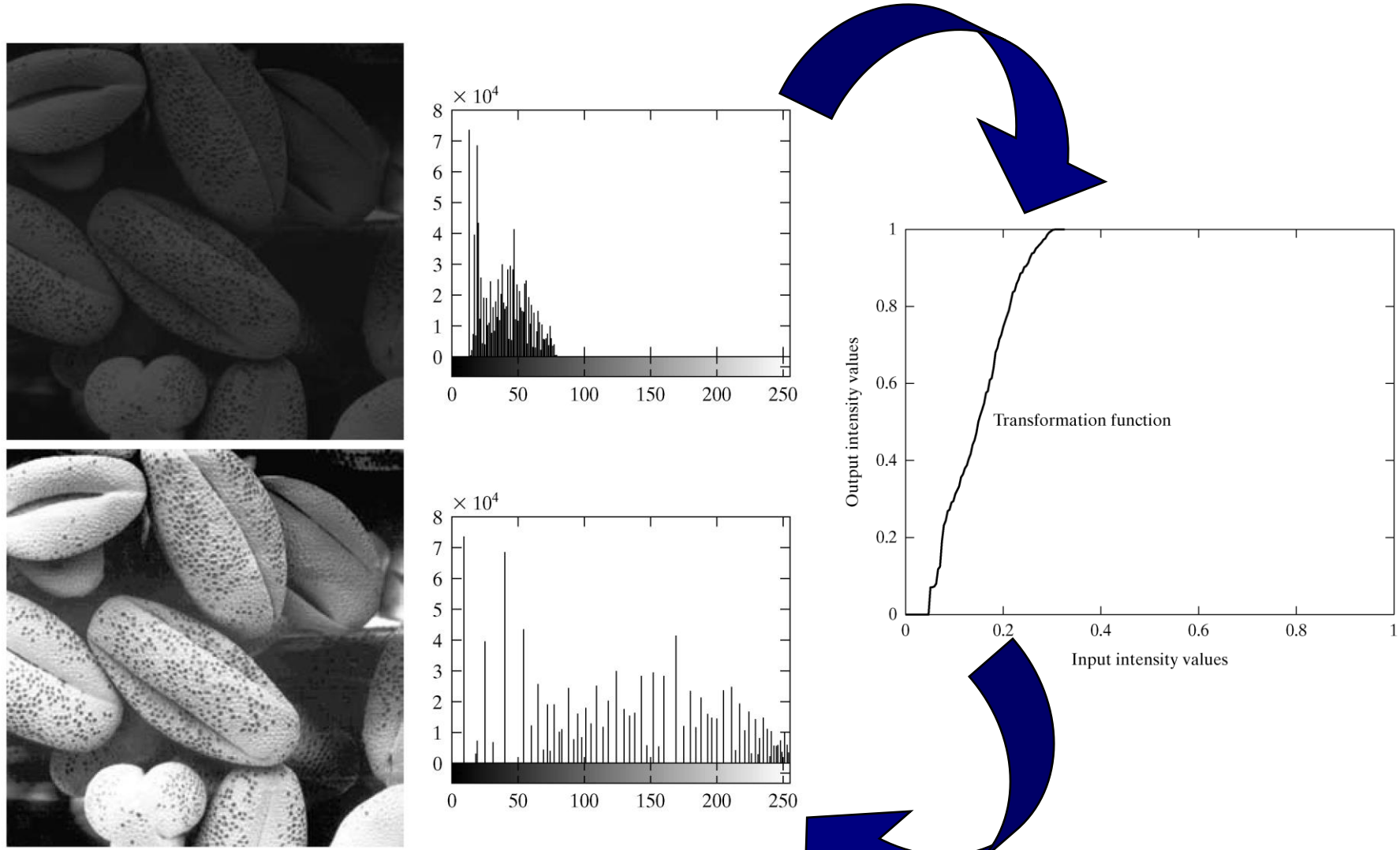
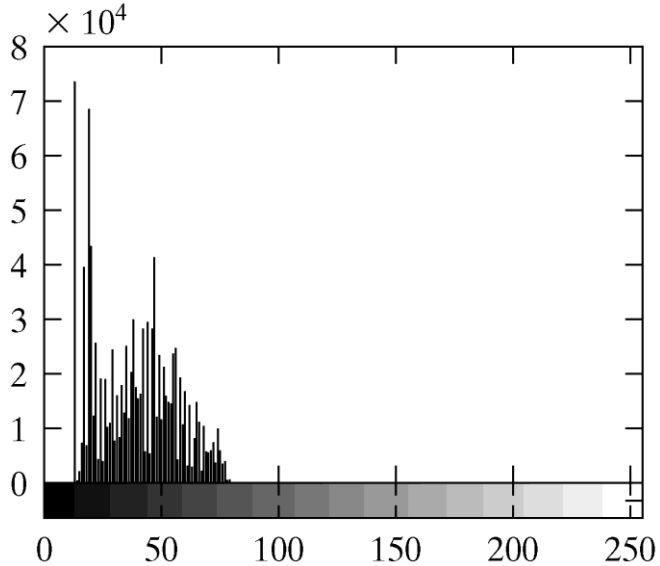


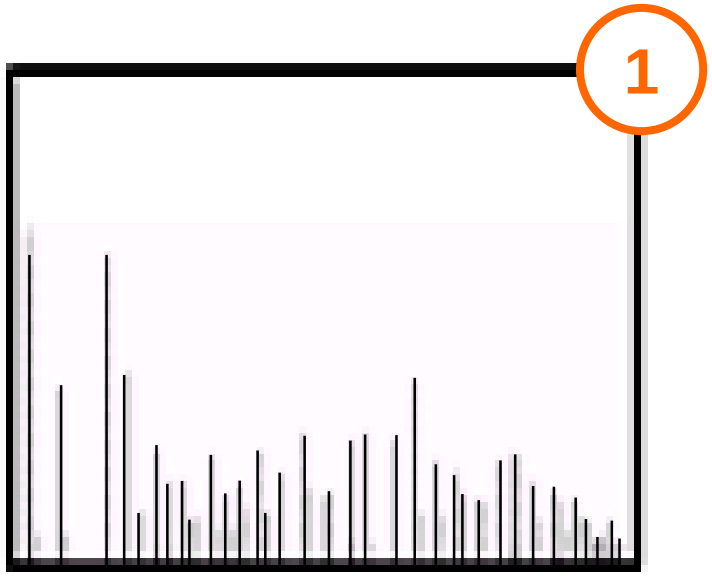
Figure: Left Column: images
Center Column: corresponding histogram equalized images.
Right Column: histogram of the images in the center column

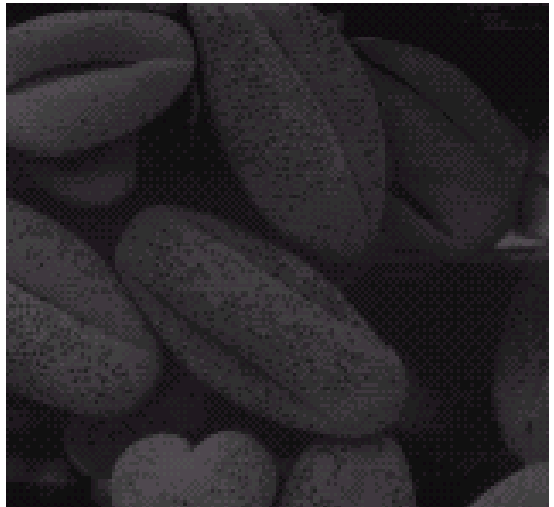
Equalization Transformation Function



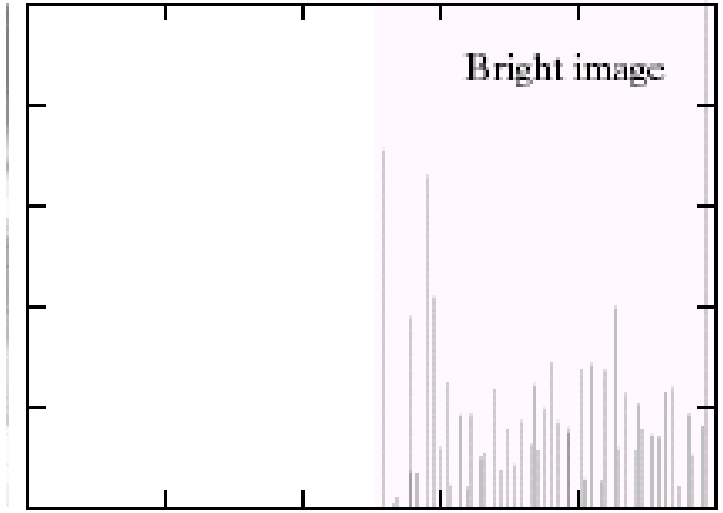
Equalization Example

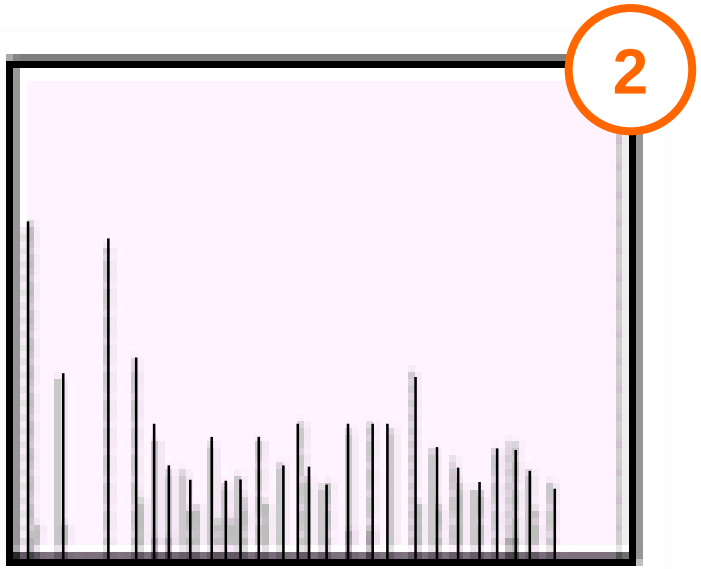




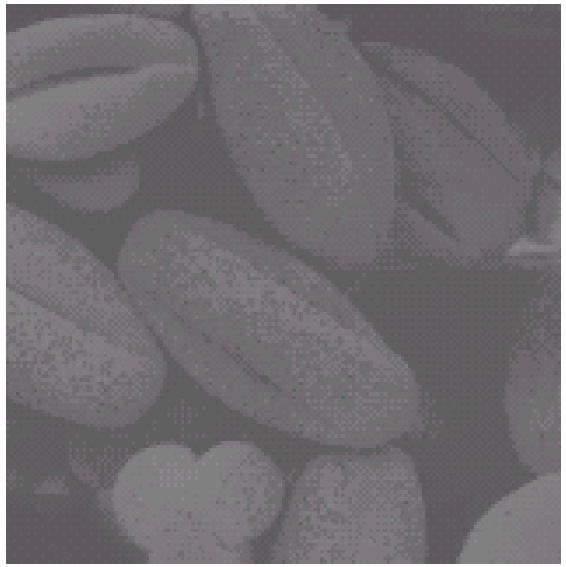


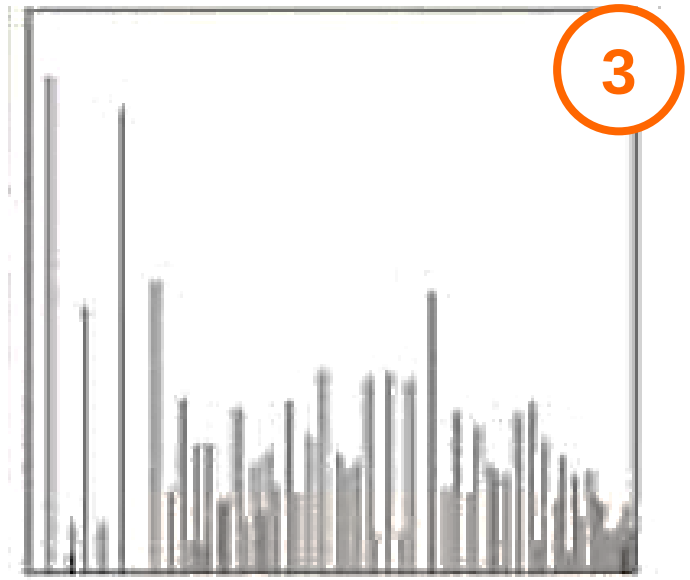
Equalization Example

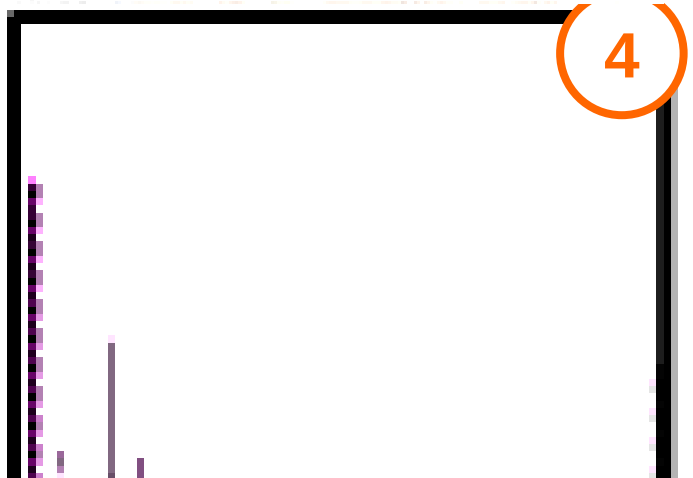




Equalization Example

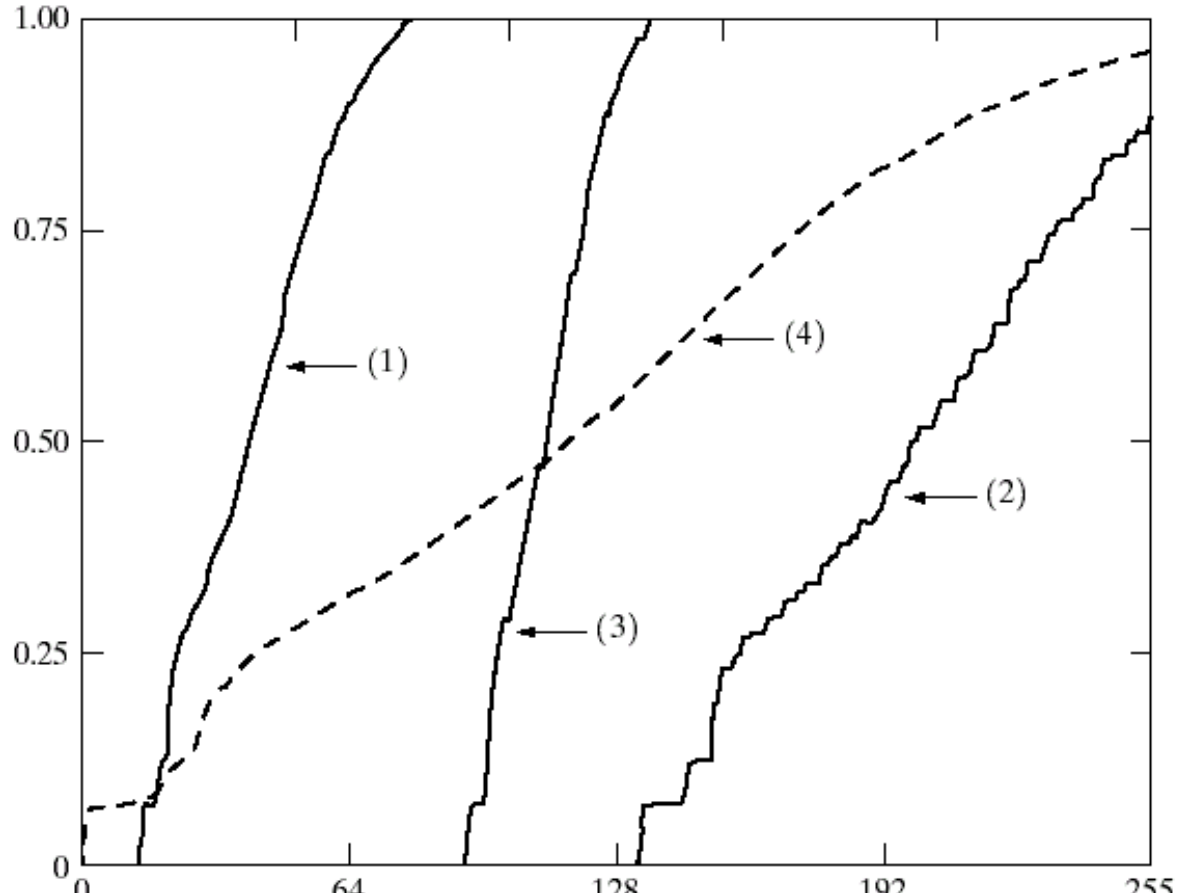






Equalization Transformation Functions

- The functions used to equalise the images in the previous examples



Numerical Examples

- Suppose that a 3-bit image (L=8) of size 64 x 64 pixels (MN = 4096) has the intensity distribution shown in table below, where the intensity levels are in integers in the range [0, L-1] = [0, 7]. Perform histogram equalization.

Gray level (r_k)	0	1	2	3	4	5	6	7
No. of pixel (n_k)	8	10	10	2	12	16	4	2

Gray level(r_k)	No. of Pixels(n_k)	PDF(n_k/N) $P_r(r_k)$	CDF	(L-1)*CDF	H_k
0	8	0.13	0.13	0.91	1
1	10	0.16	0.29	2.03	2
2	10	0.16	0.45	3.15	3
3	2	0.03	0.48	3.36	3
4	12	0.18	0.66	4.62	5
5	16	0.25	0.91	6.37	6
6	4	0.06	0.97	6.79	7
7	2	0.03	1.0	7	7
	64	1			

Drawbacks of Histogram Equalization

- Histogram equalization automatically determines a transformation function that seeks to produce an output image that has a uniform histogram.
- When automatic enhancement is desired, it is a good approach because the results from this technique are predictable and the method is simple to implement.

- Histogram equalization may not always produce desirable results, particularly if the given histogram is very narrow. It can produce false edges and regions. It can also increase image “graininess” and “patchiness”.

Spatial Filtering

Introduction

- Many image enhancement techniques are based in spatial operations. Most of the spatial operations are performed on the basis of local neighborhood of the input pixel.
- Some of the example of spatial operations for image enhancement are as below:
  a. Neighborhood processing
  b. Spatial averaging
  c. Smoothing spatial filter
  d. Zooming
  e. Low pass, high pass and band pass

Definition of Spatial Filtering

- Use of spatial filters (also called *spatial masks*, *kernels*, *templates*, *and windows*) for filtering is called spatial filtering.
- It may be linear or non linear.
- Used of image enhancement.

- A spatial filter consists of (1) *neighborhood*, (typically a small rectangle), and (2) *a predefined operation* that is performed on the image pixels encompassed by the neighborhood.
- Filtering creates a new pixel with coordinates equal to the coordinates of the center of the neighborhood, and whose value is the result of the filtering operation. A processed (filtered) image is generated as the center of the filter visits each pixel in the input image.

Linear Filters

- **Low-pass**: attenuate (or eliminate) high frequency components such as characterized by edges and sharp details in an image
    - Net effect is image blurring
- **High-pass**: attenuate (or eliminate) low frequency components such as slowly varying characteristics
    - Net effect is a sharpening of edges and other details
- **Band-pass**: attenuate (or eliminate) a given frequency range
    - Used primarily for image restoration(are of little interest for image enhancement)

Non Linear

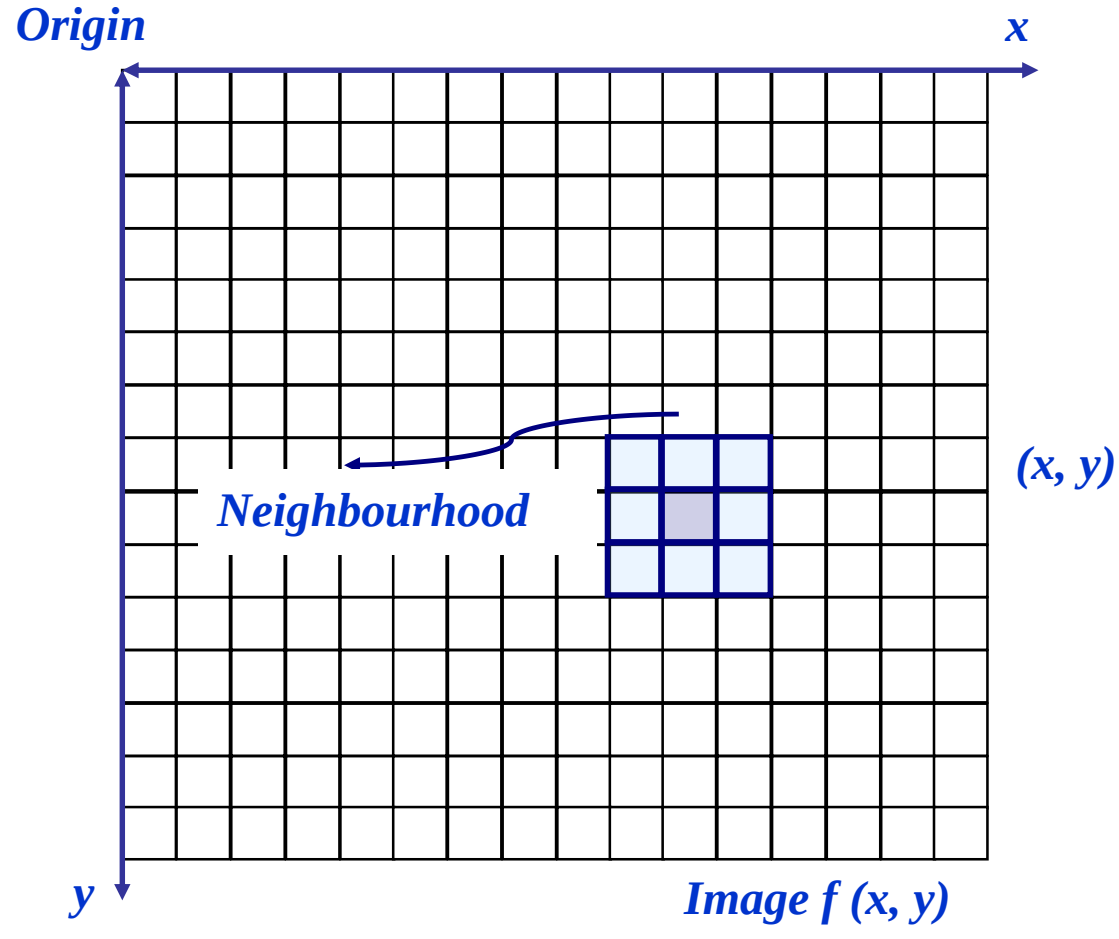
- Nonlinear spatial filters also operate on neighbourhoods
- Their operation is based directly on pixel values in the neighbourhood under consideration.
  - They do not explicitly use coefficient values as in the linear spatial filters.
- Example nonlinear spatial filters
  - **Median filter**: Computes the median gray-level value of the neighbourhood. Used for noise reduction.
  - **Max filter**: Used to find the brightest points in an image

    $R=\max\{z_k \mid k=1,2,\ldots,9\}$
  - **Min filter**: Used to find the dimmest points in an image

    $R=\min\{z_k \mid k=1,2,\ldots,9\}$

Neighborhood Processing

- Neighborhood processing is the image processing technique in which the resulting value for a pixel at coordinates (x, y) (also called *reference pixel*) is a function of the original pixel value at that point as well as its neighbors.
- Neighborhoods are mostly a rectangle around a central pixel.
- Any size rectangle and any shape filter are possible. (size of neighbor 3 x 3, 5 x 5, 7 x 7 etc.)



Some Neighborhood Operations

Min: Set the pixel value to the minimum in the neighbourhood.

Max: Set the pixel value to the maximum in the neighbourhood.

Median: The median value of a set of numbers is the midpoint value in that set (e.g. from the set [1, 7, 15, 18, 24] 15 is the median). Sometimes the median works better than the average.

The Spatial Filtering Process

- At any point (x, y) in the image, the response $g(x, y)$, of the filter is the sum of products of the filter coefficients and the image pixels encompassed by the filter:

$$g(x, y) = w(-1, -1)\, f(x-1, y-1) + w(-1, 0)\, f(x-1, y) + \ldots\ldots + w(0, 0)\, f(x, y) + w(1, 1)\, f(x+1, y+1)$$

- The center of coefficient of the filter , $w(0, 0)$, aligns with the pixel at location (x, y)
- For a mast of $m \times n$, we assume $m = 2a+1$ and $n = 2b + 1$ where a and b are some positive integers.
- In general, linear spatial filtering of an image of size $M \times N$ with a filter of size $m \times n$ is given by the expression:

$$g(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t)\, f(x+s, y+t)$$

where x and y are varied so that each pixel in w visits every pixels in f.

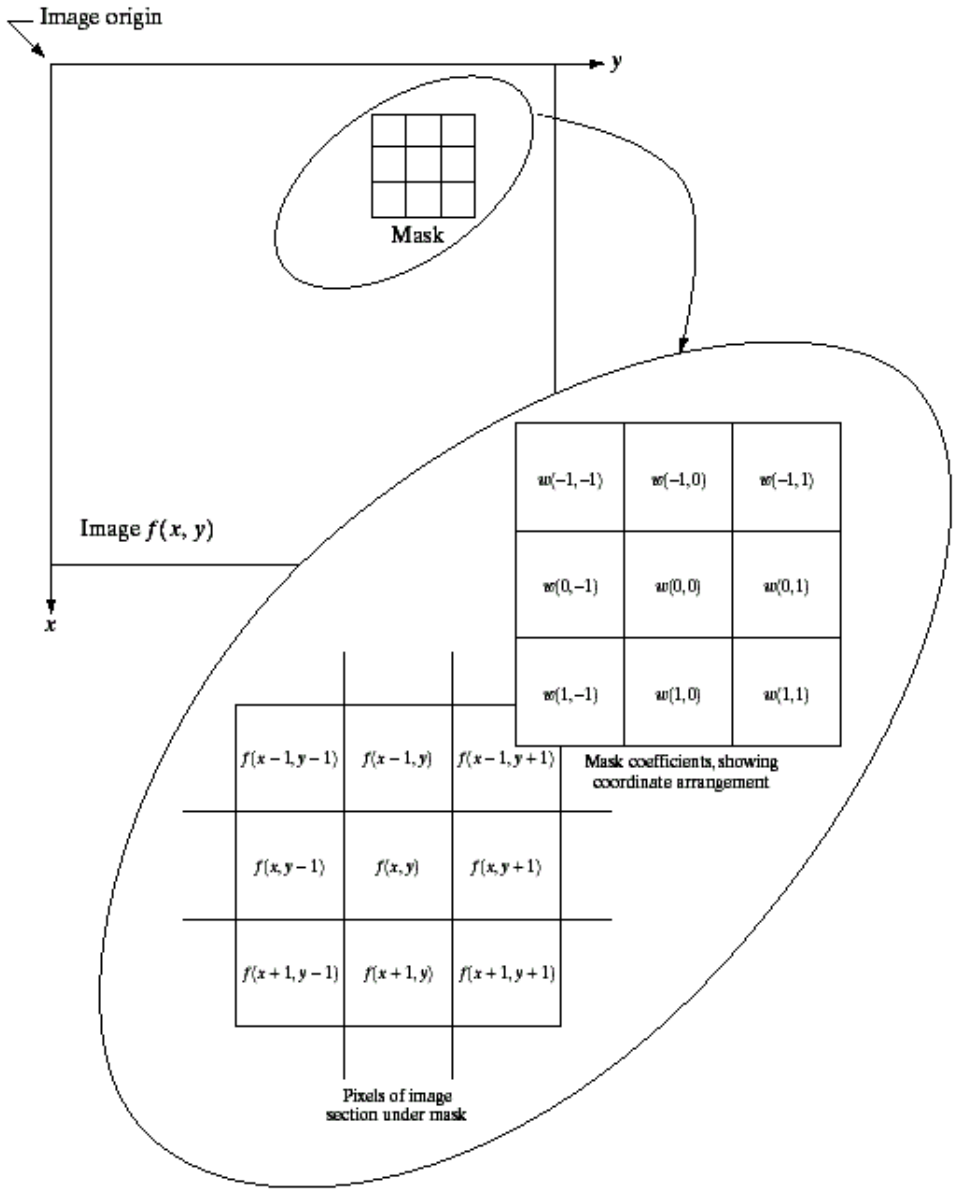
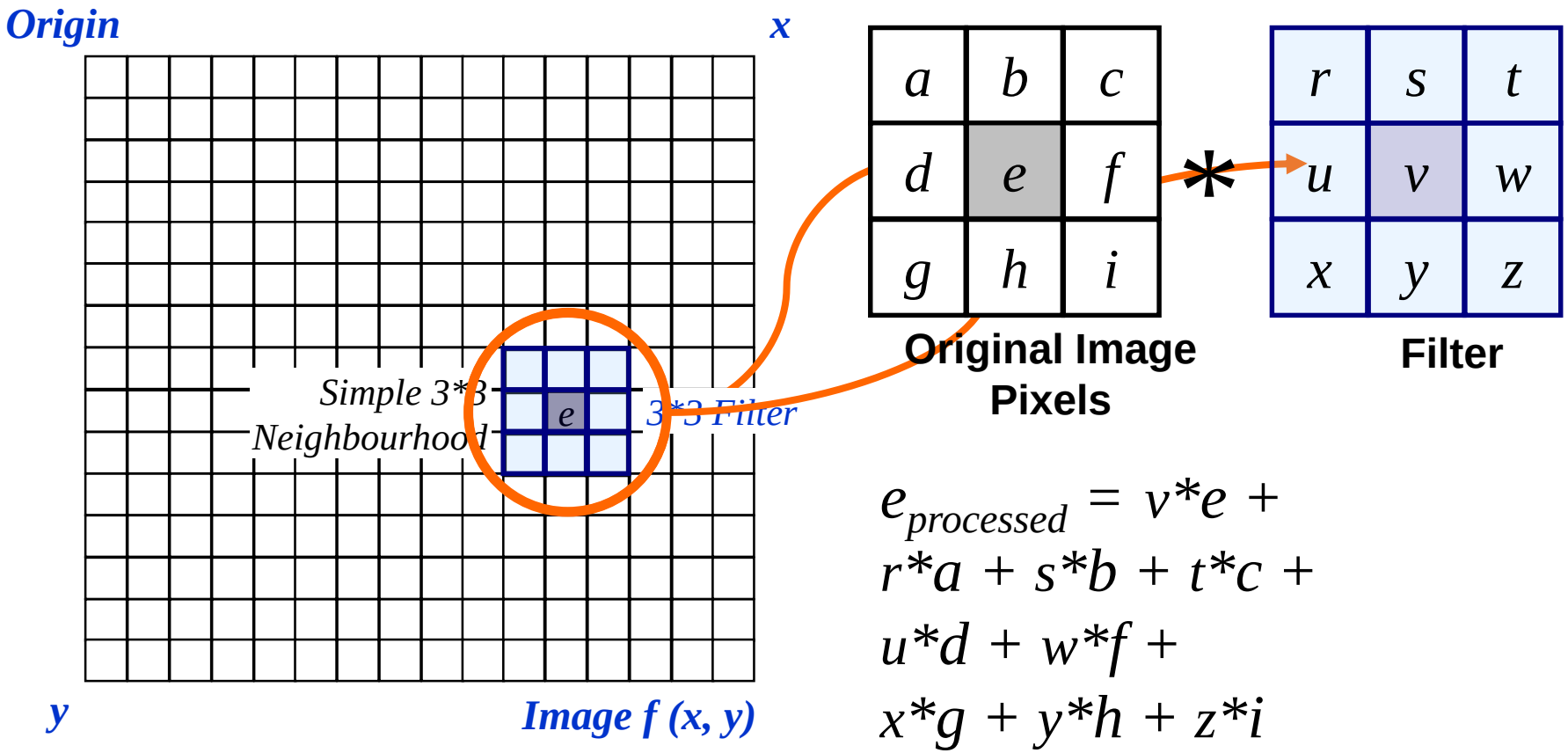


Figure: The mechanics of linear spatial filtering

The Spatial Filtering Process



$$e_{processed} = v*e + r*a + s*b + t*c + u*d + w*f + x*g + y*h + z*i$$

The above is repeated for every pixel in the original image to generate the filtered image.

Vector Representation of Linear Filtering

- Written as sum of products as:

$$R = w_1 z_1 + w_2 z_2 + \ldots\ldots\ldots w_{mn} z_{mn}$$

- Where *w*s are the coefficients of an *m x n* filter and the *z*s are the corresponding image intensities encompassed by the filter.
- For example: for given 3 x 3 masks with coefficients labeled as follows, the R value is given by:

W_1	W_2	W_3
W_4	W_5	W_6
W_7	W_8	w_9

Figure: A representation of 3 x 3 filter mask.

$$R = w_1 z_1 + w_2 z_2 + \ldots\ldots\ldots + w_9 z_9$$

$$= \sum_{k=1}^{9} w_k z_k$$

$$= w^T z$$

Generating Spatial Filter Masks

- Generating an *m x n* linear spatial filter requires that we specify *mn* mask coefficients. In turn, these coefficients are based on what the filter is supposed to do.

For example:

- Suppose we want to replace the pixels in an image by the average intensity of a *3 x 3* neighborhood centered on those pixels.
- The average value at any location (*x ,y*) in the image is the sum of the nine intensity values in the *3 x 3* neighborhood centered on (*x, y*) divided by 9. letting z_i, *i = 1, 2, 3, ……., 8, 9*, denote these intensities, the average is

$$R = \frac{1}{9}\sum_{i=1}^{9} z_i$$

- Here the coefficient of mask is $w_i = 1/9$, for *i = 1 , 2, ……, 9*
- In other words, a linear filtering operation with 3 x 3 mask whose coefficients are 1/9 implements the desired averaging.

Smoothing Spatial Filters

- Used for blurring and for noise reduction.
- Blurring is used in preprocessing tasks, such as removal of small details from an image prior to (large) object extraction, and bridging of small gaps in lines or curves.

- *The output (response) of a smoothing, linear spatial filter is simply the average of the pixels contained in the neighborhood of the filter mask*. These filters sometimes are called *averaging filters*. They are also referred to a *low-pass filters*.

Idea behind smoothing filters

- The idea behind smoothing filters is straightforward. By replacing the value of every pixel in an image by the average of the intensity levels in the neighborhood defined by the filter mask, this process results in an image with reduced “sharp” transitions in intensities. Because random noise typically consists of sharp transitions in intensity levels, the most obvious application of smoothing is noise reduction.
- However, edges, (which almost always are desirable features of an image) also characterized by sharp intensity transitions, so averaging filters have the undesirable side effect that they blur edges.

Smoothing Spatial Filters

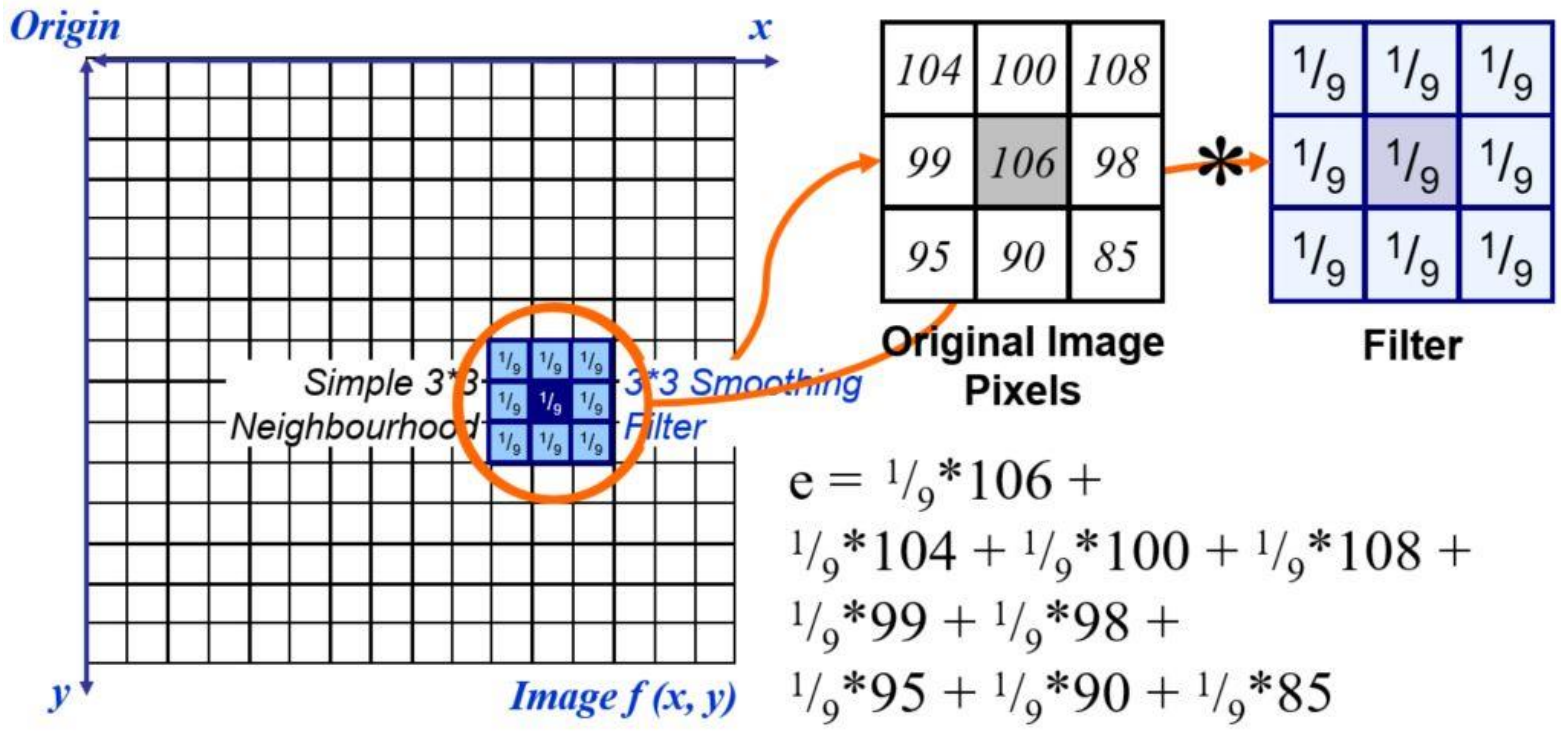
Example:

- For a 3x3 mask, the simplest arrangement is to have all the coefficient values equal to one (neighborhood averaging)
  - The response would be the sum of all gray levels for the nine pixels in the mask.
  - This could cause the value of R to be out of the valid gray-level range.
  - The solution is to scale the result by dividing by 9

$$\frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \end{bmatrix}$$

Figure: A 3 x 3 smoothing (averaging filter mask)

Smoothing Spatial Filtering (contd…)

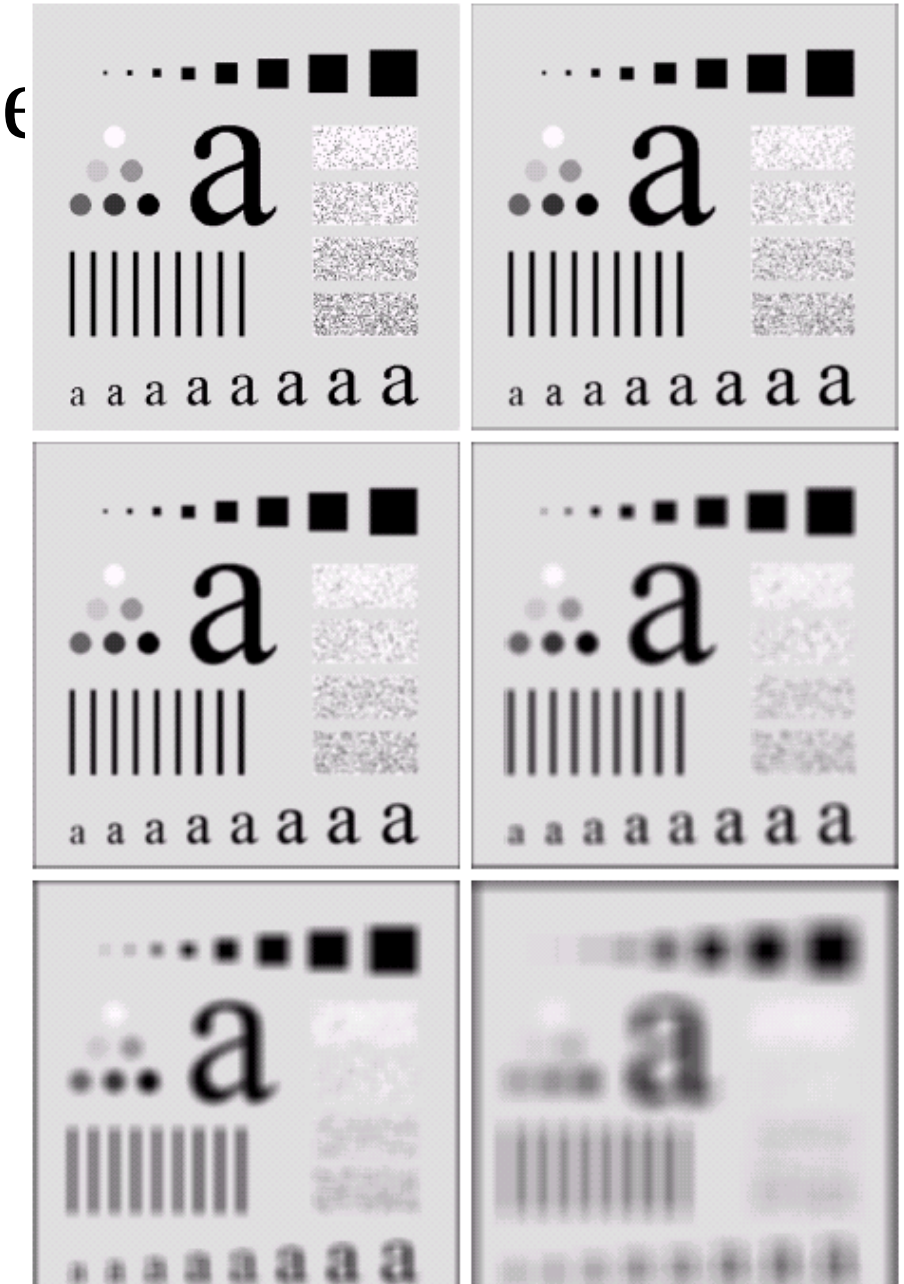


$$e = \tfrac{1}{9}*106 + \tfrac{1}{9}*104 + \tfrac{1}{9}*100 + \tfrac{1}{9}*108 + \tfrac{1}{9}*99 + \tfrac{1}{9}*98 + \tfrac{1}{9}*95 + \tfrac{1}{9}*90 + \tfrac{1}{9}*85 = 98.3333$$

The above is repeated for every pixel in the original image to generate the smoothed image

Image Smoothing Example

- The image at the top left is an original image of size 500 x 500 pixels.
- The subsequent images show the image after filtering with an averaging filter of increasing sized 3, 5, 9, 15 and 35.
- Notice how detail begins to disappear.



Weighted Smoothing Filters

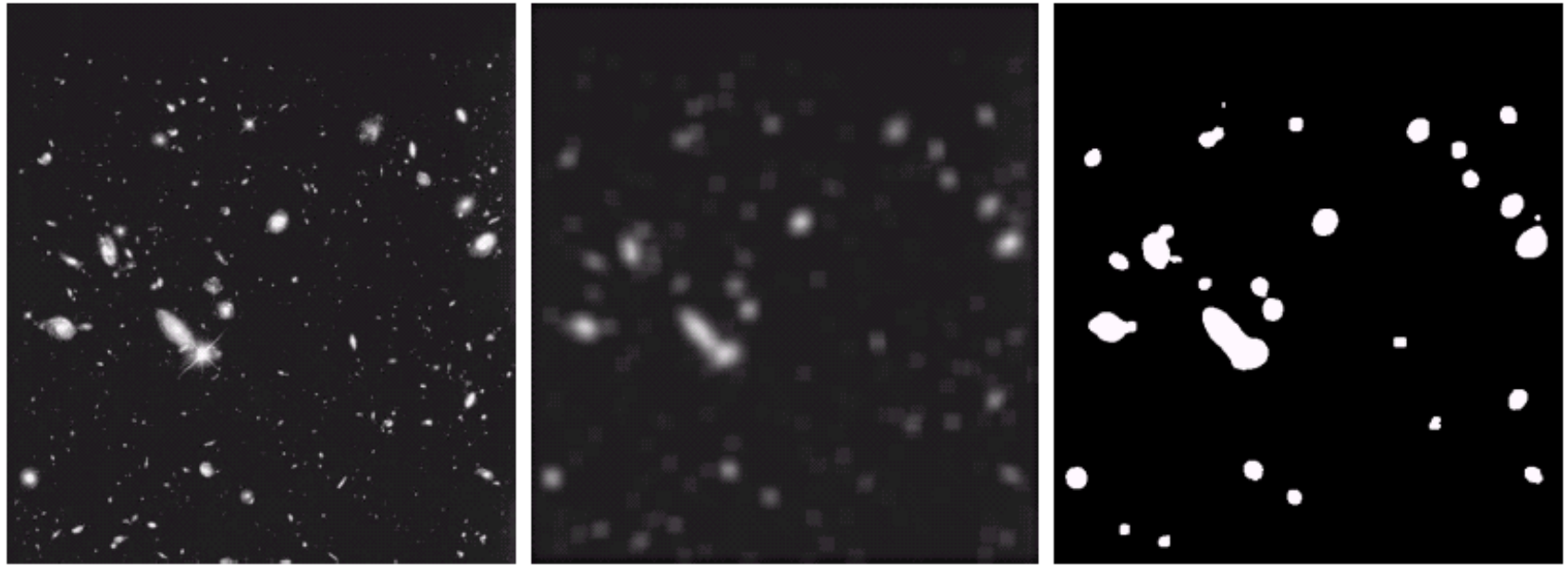
- More effective smoothing filters can be generated by allowing different pixels in the neighbor different weights in the averaging function.
- Pixels closer to the central pixel are more important.
- Often referred to as a *weighted averaging*.

$^1/_{16}$	$^2/_{16}$	$^1/_{16}$
$^2/_{16}$	$^4/_{16}$	$^2/_{16}$
$^1/_{16}$	$^2/_{16}$	$^1/_{16}$

Weighted averaging filter

Problem with averaging filter

- By smoothing the original image we get rid of lots of the finer detail which leaves only the gross features for thresholding.



(a) (b) (c)

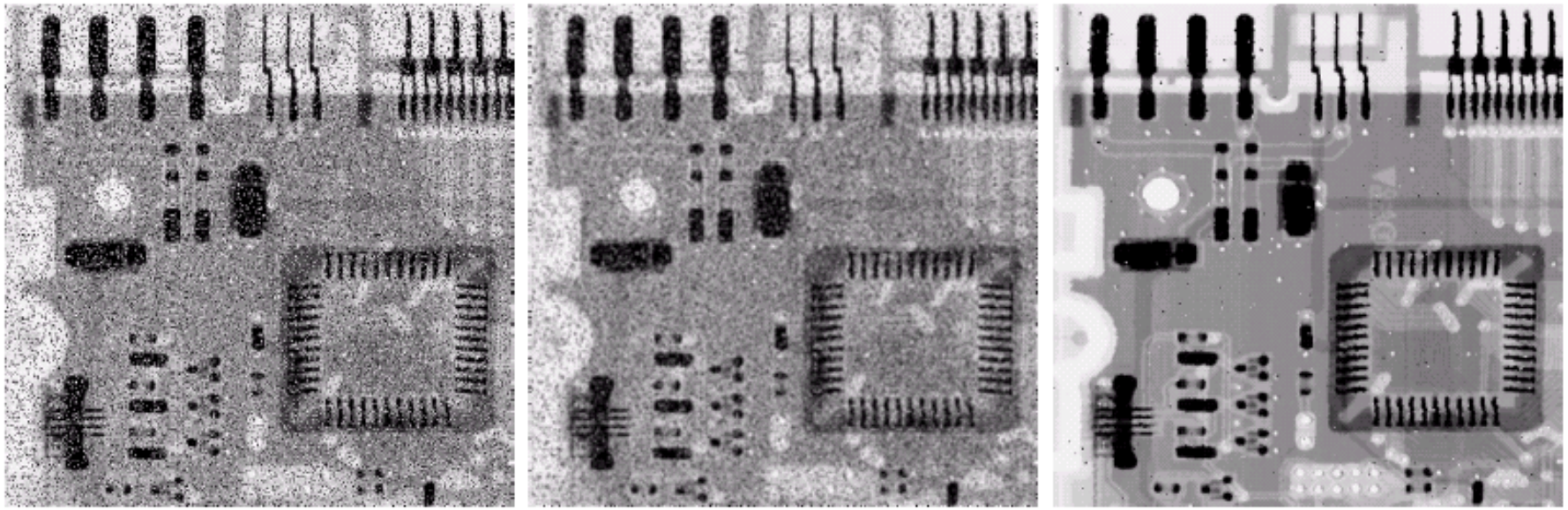
Figure: (a) Image of size 528 x 485 pixels from the Hubble Space Telescope, (b) Image filtered

Smoothing Filters: Median Filtering

One problem with the lowpass filter is it blurs edges and other sharp details.

- If the intent is to achieve noise reduction, one approach can be to use median filtering.
  - The value of each pixel is replaced by the median pixel value in the neighbourhood (as opposed to the average)
  - Particularly effective when the noise consists of strong, spike like components and edge sharpness is to be preserved
- The median m of a set of values is such that half of the values are greater than m and half are less than m
- To implement, sort the pixel values in the neighbourhood, choose the median and assign this value to the pixel of interest
- Forces pixels with distinct intensities to be more like their neighbours

Averaging Filter Vs. Median Filter Example



Original Image With Noise | **Image After Averaging Filter** | **Image After Median Filter**

(a) **(b)** **(c)**

Figure: (a) X-ray image of circuit corrupted by salt-and-pepper noise, (b) Noise reduction with a 3 x 3 averaging mask, (c) Noise reduction with a 3 x 3 median filter.

- Filtering is often used to remove noise from images.

Smoothing Spatial Filter: Median Filter

- The median filter is a also a type of spatial filter but it replaces the central value in the window with median of all the pixel values in the window.
- The kernel or template or mask is usually square but can be initiated in different window.

Example:

Unfiltered values in order like

0, 2, 3, 4, 6, 10, 15, 19, 97

Now,

Output value $= \frac{n+1}{2}$ th value

Or median value $= \frac{9+1}{2}$ th value

$= 5^{th}$ value

$= 6$

6	**2**	**0**
3	97	4
19	15	10

Before Filtering

6	**2**	**0**
3	6	4
19	15	10

After Filtering

The center value 97 is replaced by the median value of all nine values i.e. 6

Smoothing Spatial Filter: Mean Filter

- Mean filter is a spatial filter that replaces the center value in the window with the average of all pixel values in the window.
- Here, the window can be of any shape but can be initiated as a square mask. Therefore, average of smoothing filter shows the properties of low pass filter. So it reduces the noise from an image.

Example:

Unfiltered values in order like:

5, 3, 6, 2, 1, 9, 8, 4, 7

Now,

Output value or mean value $= \frac{\sum x}{N}$

$= \frac{5+3+6+2+1+9+8+4+7}{9}$

$= \frac{45}{9}$

$= 5$

5	**3**	**6**
2	1	9
8	4	7

Before Filtering

5	**3**	**6**
2	5	9
8	4	7

After Filtering

The center value 97 is replaced by the mean value of all nine values i.e. 5

Sharpening Filters (High Pass Filter)

- The principal objective *to highlight fine details in an image or to enhance detail that has been blurred*, either in error or as a natural effect of a particular method of image acquisition.
- Sharpening is *accomplished by spatial differentiation*.
- Fundamentally, the strength of the response of a derivative operator is proportional to the degree of discontinuity of the image at the point at which the operator is applied.
- Thus, image differentiation enhances edges and other discontinuities (such as noise) and deemphasizes areas with slowly varying gray-level values.

Sharpening Filters (High Pass Filter) (contd…)

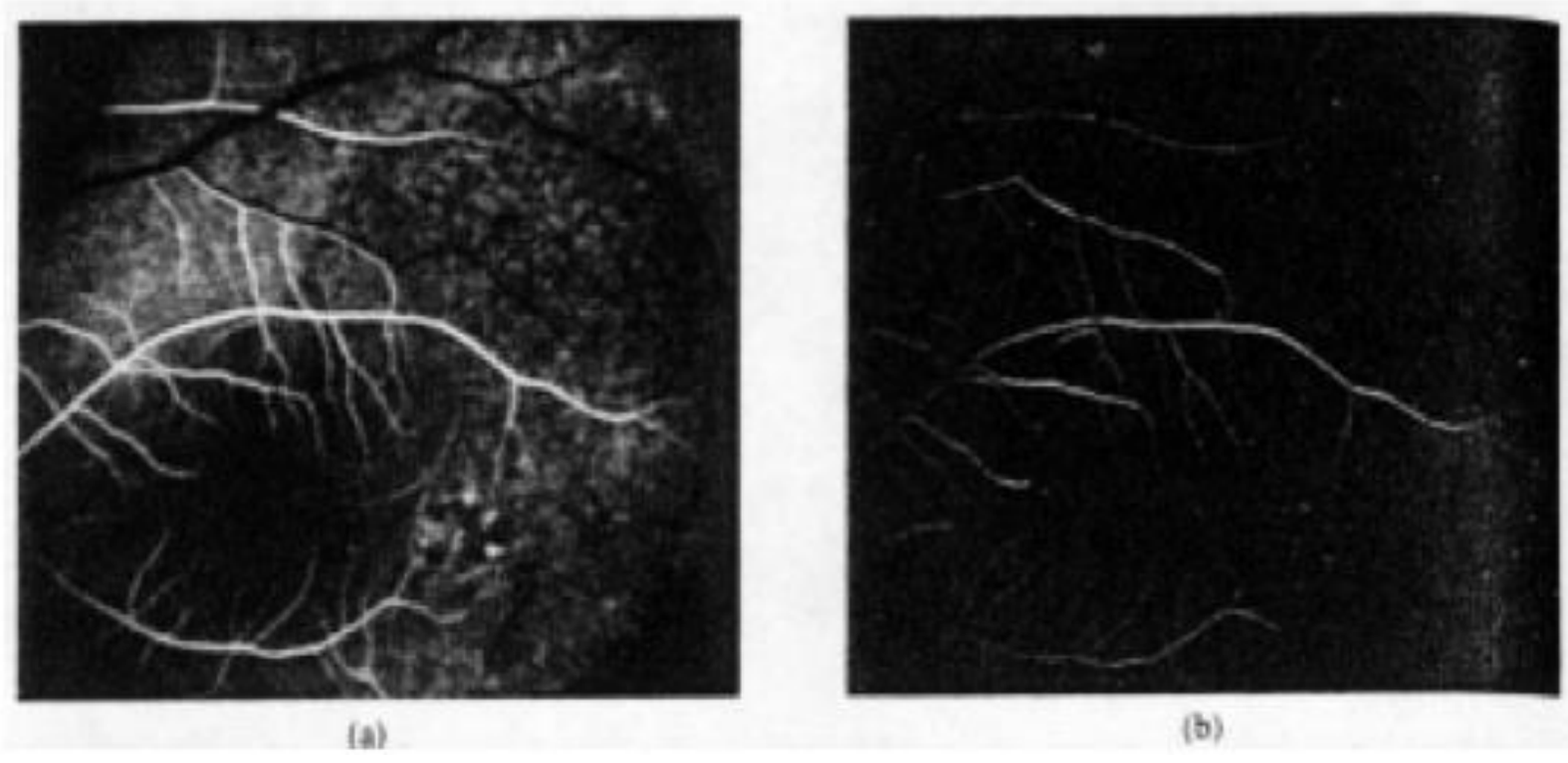
- The shape of the impulse response needed to implement a high-pass (sharpening) filter indicates the filter should have positive coefficients near its center and negative coefficients in the outer periphery.
- For a 3x3 mask, the simplest arrangement is to have the center coefficient positive and all others negative.

$$\frac{1}{9}\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Sharpening Filters (High Pass Filter) (contd…)

- Note the sum of the coefficients is zero
  - When the mask is over a constant or slowly varying region the output is zero or very small
    - This filter eliminates the zero-frequency term.
    - Eliminating this term reduces the average gray-level value in the image to zero (will reduce the global contrast of the image)
    - Result will be a some what edge-enhanced image over a dark background.
- Reducing the average gray-level value to zero implies some negative gray-levels.
  - The output should be scaled back into an appropriate range [0, L-1]

Sharpening Filters (High Pass Filter) (contd…)



Sharpening Filters: High-Boost Filter

A high-pass filter may be computed as:

High-pass = Original - Lowpass

- Multiplying the original by an amplification factor yields a high-boost or high-frequency-emphasis filter

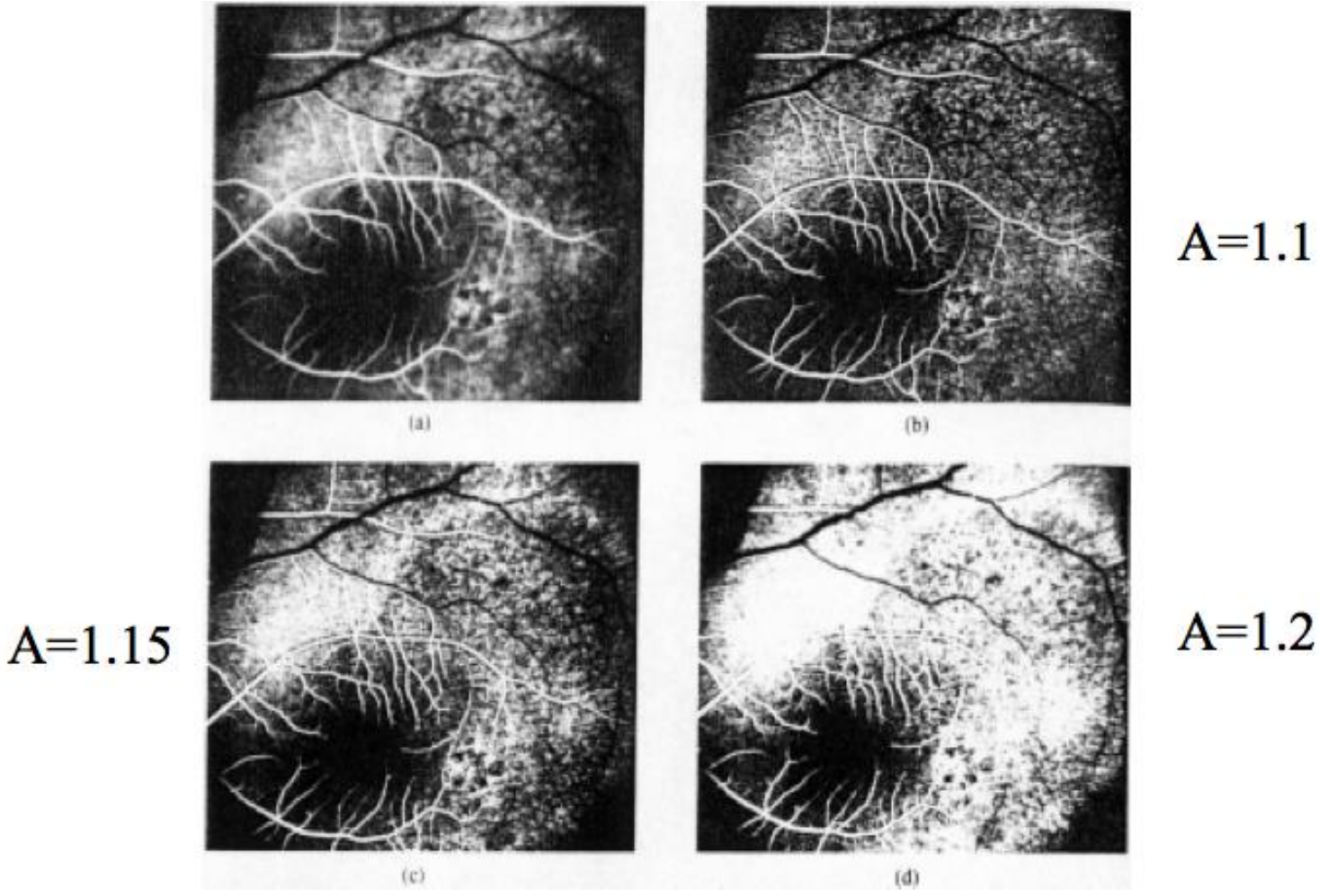
$$\text{High-boost} = A(\text{Original}) - \text{Lowpass}$$
$$= (A - 1)(\text{Original}) + \text{Original} - \text{Lowpass}$$
$$= (A - 1)(\text{Original}) + \text{High-pass}$$

  – If A>1, part of the original image is added to the high-pass result (partially restoring low frequency components).

  – Result looks more like the original image with a relative degree of edge enhancement that depends on the value of A.

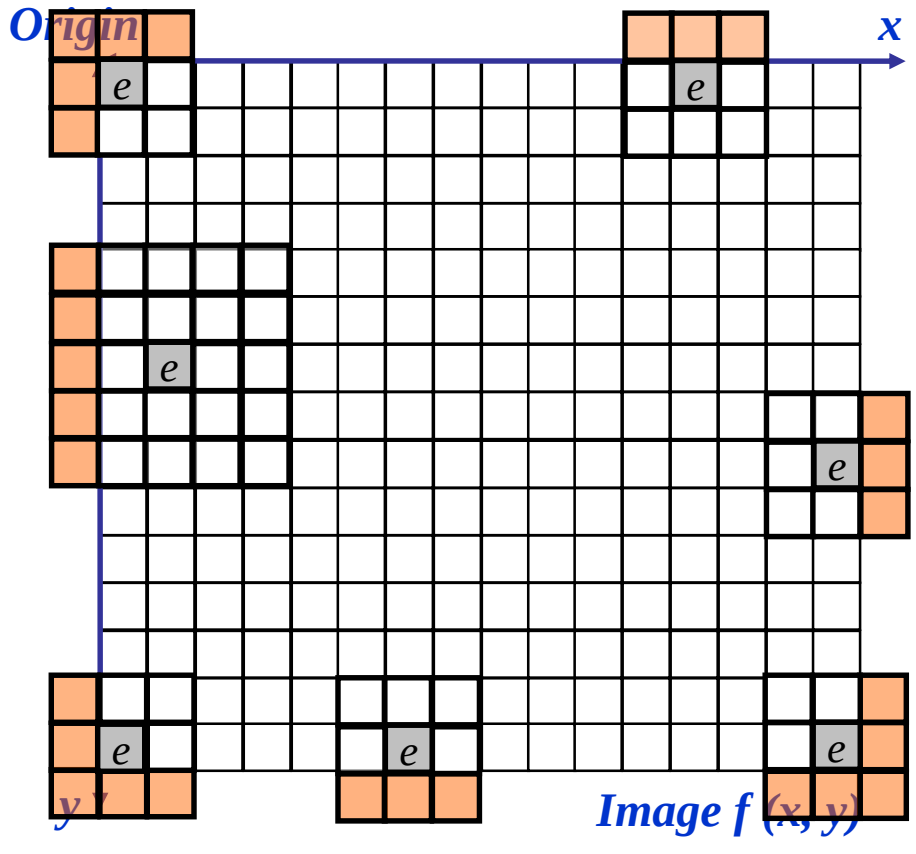
  – May be implemented with the center coefficient value $w=9A-1$ $(A\geq 1)$

Sharpening Filters: High-Boost Filter



Strange Things Happen at The Edges

- At the edges of an image, we are missing pixels to form a neighborhood.



Strange Things Happen at The Edges (contd…)

- There are a few approaches to dealing with missing edge pixels:

1. Omit missing pixels
   - Only works with some filters
   - Can add extra code and slow down processing
2. Pad the image
   - Typically with either all white or all black pixels
3. Replicate border pixels
4. Truncate the image
5. Allow pixels wrap around the image
6. Can cause some strange image artifacts.

Zooming

- It is one of the applications of image enhancement where spatial domain neighborhood operation is used.
- Zooming is the process of making an image larger. It involves two steps:
  1. Creating new pixel location
  2. Assigning gray level of those new location
- Different types of zooming technique can be used to zoom. They are:
  1. Replication
  2. Linear interpolation

1. Replication

- In replication, we simply replicate each pixel and then replicate row.

Let us consider an image as:

1	**2**	**3**	**4**
5	6	7	8
9	8	6	7
0	1	2	3

As we start from the first row, first we replicate each pixel and then replicate each row.

Now,

The first row looks as: [1 1 2 2 3 3 4 4]

1. Replication (contd…)

- We will now replicate this row then it looks as:

$$\begin{bmatrix} 1 & 1 & 2 & 2 & 3 & 3 & 4 & 4 \\ 1 & 1 & 2 & 2 & 3 & 3 & 4 & 4 \end{bmatrix}$$

Now,

Performing replication operation on an entire image then we obtain output image as:

Here, 4x4 image is zoomed to 8x8 image.
This method can be repeated to get larger image but no new data is added

1	**1**	**2**	**2**	**3**	**3**	**4**	**4**
1	1	2	2	3	3	4	4
5	5	6	6	7	7	8	8
5	5	6	6	7	7	8	8
9	9	8	8	6	6	7	7
9	9	8	8	6	6	7	7
0	0	1	1	2	2	3	3
0	0	1	1	2	2	3	3

Zooming by zero-interlace technique

Step (1): Adding zero after every pixel of the first row then we obtain as:

[1 0 2 0 3 0 4 0]

Step (2) : Adding zero at every column

Now,

Insert row of zero and then this is know as zero-interlacing

1 0 2 0 3 0 4 0

0 0 0 0 0 0 0 0

Step (3) :

Now,

Performing zero- interlace technique on an entire image then we obtain output image as:

1	**0**	**2**	**0**	**3**	**0**	**4**	**0**
0	0	0	0	0	0	0	0
5	0	6	0	7	0	8	0
0	0	0	0	0	0	0	0
9	0	8	0	7	0	6	0
0	0	0	0	0	0	0	0
0	0	1	0	2	0	3	0
0	0	0	0	0	0	0	0

Figure: 8x8 final pseudo image

Hence, this image is known as zero-interlace image.

2. Linear Interpolation

- In this method, instead of replicating each pixel, average of the two adjacent pixel along the row is taken and placed between the two pixels.
- Accordingly, same operation is then performed along the column.
- Linear interpolation is also derived from the zero-interlace technique.

Step (1) : zero- interlace

1	0	2	0	3	0	4	0
0	0	0	0	0	0	0	0
5	0	6	0	7	0	8	0
0	0	0	0	0	0	0	0
9	0	8	0	7	0	6	0
0	0	0	0	0	0	0	0
0	0	1	0	2	0	3	0
0	0	0	0	0	0	0	0

Step (2) : Interpolate row

1	1.5	2	2.5	3	3.5	4	2
0	0	0	0	0	0	0	0
3	3.5	4	4.5	5	5.5	6	3
0	0	0	0	0	0	0	0
9	8.5	8	7	6.5	6	7	3.5
0	0	0	0	0	0	0	0
0	0.5	1	1.5	2	2.5	3	3.5
0	0	0	0	0	0	0	0

Step (3) : Interpolate Column

1	1.5	2	2.5	3	3.5	4	2
3	3.5	4	4.5	5	5.5	6	3
5	5.5	6	6.5	7	7.5	8	4
7	7	7	6.25	6.5	7	7.5	3.25
9	8.5	8	7	6	6.5	7	3.5
4.5	4.5	4.5	4.5	4	4.5	5	2.5
0	0.5	1	1.5	2	2.5	3	1.5
0	0.25	0.5	0.75	1	1.25	1.3	0.7